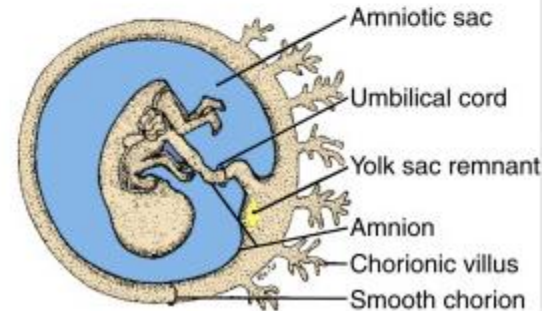
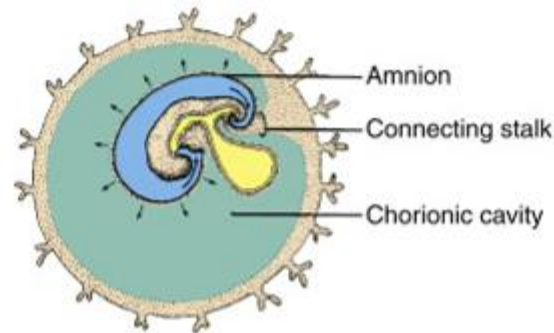
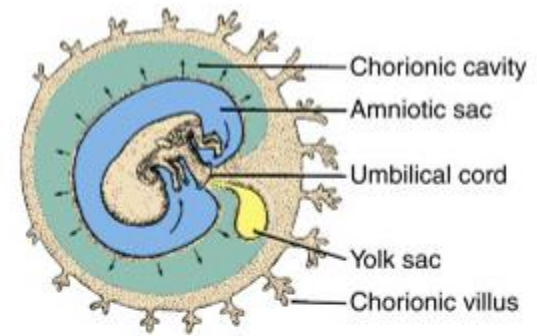
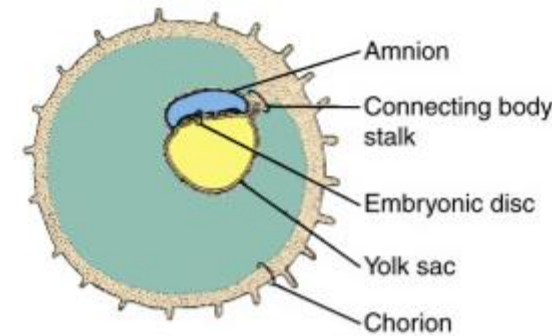


Extra-embryonic (Fetal)
Membranes
and
Placenta

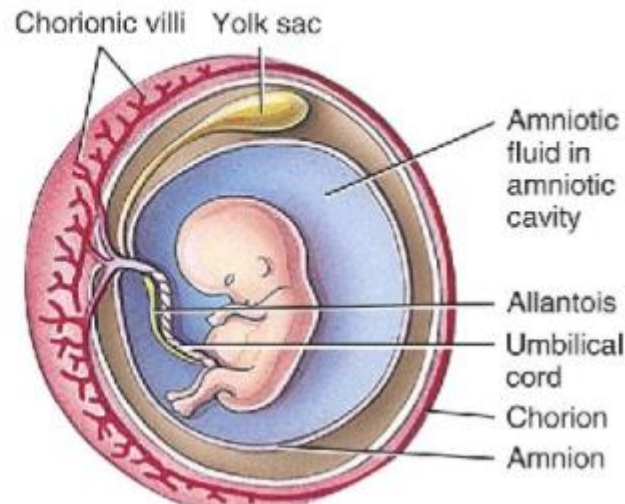
1. Extra-Embryonic membrane (Fetal membranes)

- a. Chorion
- b. Amnion
- c. Yolk sac
- d. Allantois
- e. Umbilical cord



□ 4 extraembryonic membranes:

- Amnion
- Yolk sac
- Allantois
- Chorion

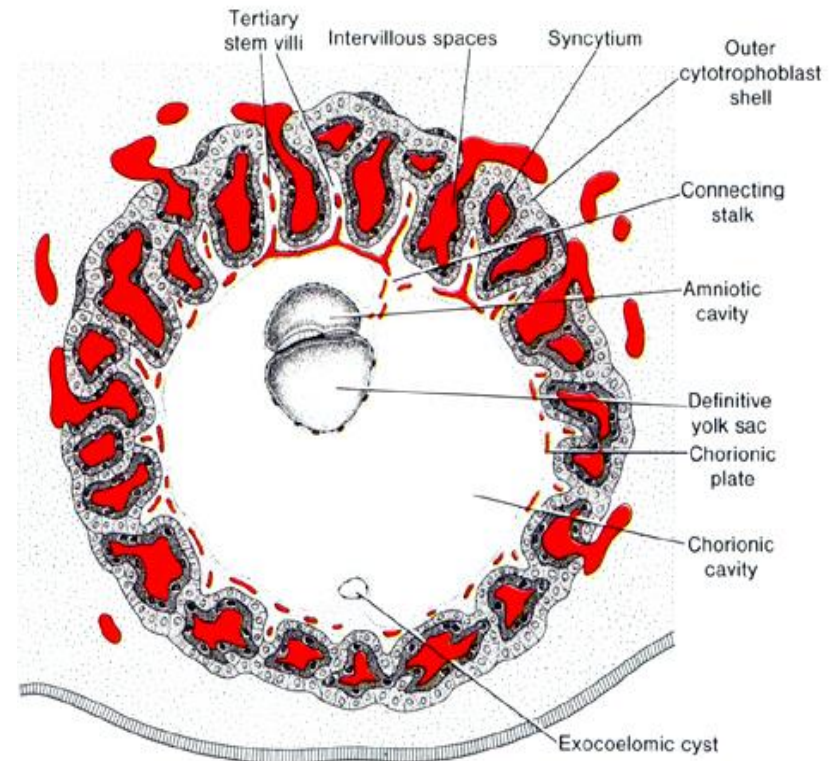
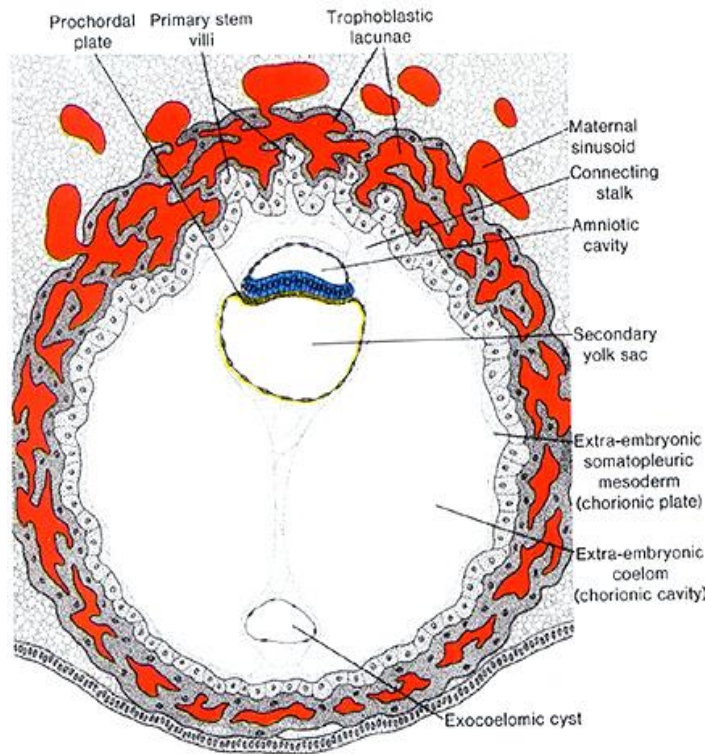


1. Chorionic:

- is a double-layered membrane formed by the trophoblast and the extra-embryonic mesoderm
- **consists of two layers:** an outer formed by the trophoblast, and an inner formed by the somatic mesoderm.

Formation:

- Formed of syncytiotrophoblast, cytotrophoblast and extraembryonic mesoderm with villi.



Development of villi:

- **develop in three stages:**

- **In the primary stage**

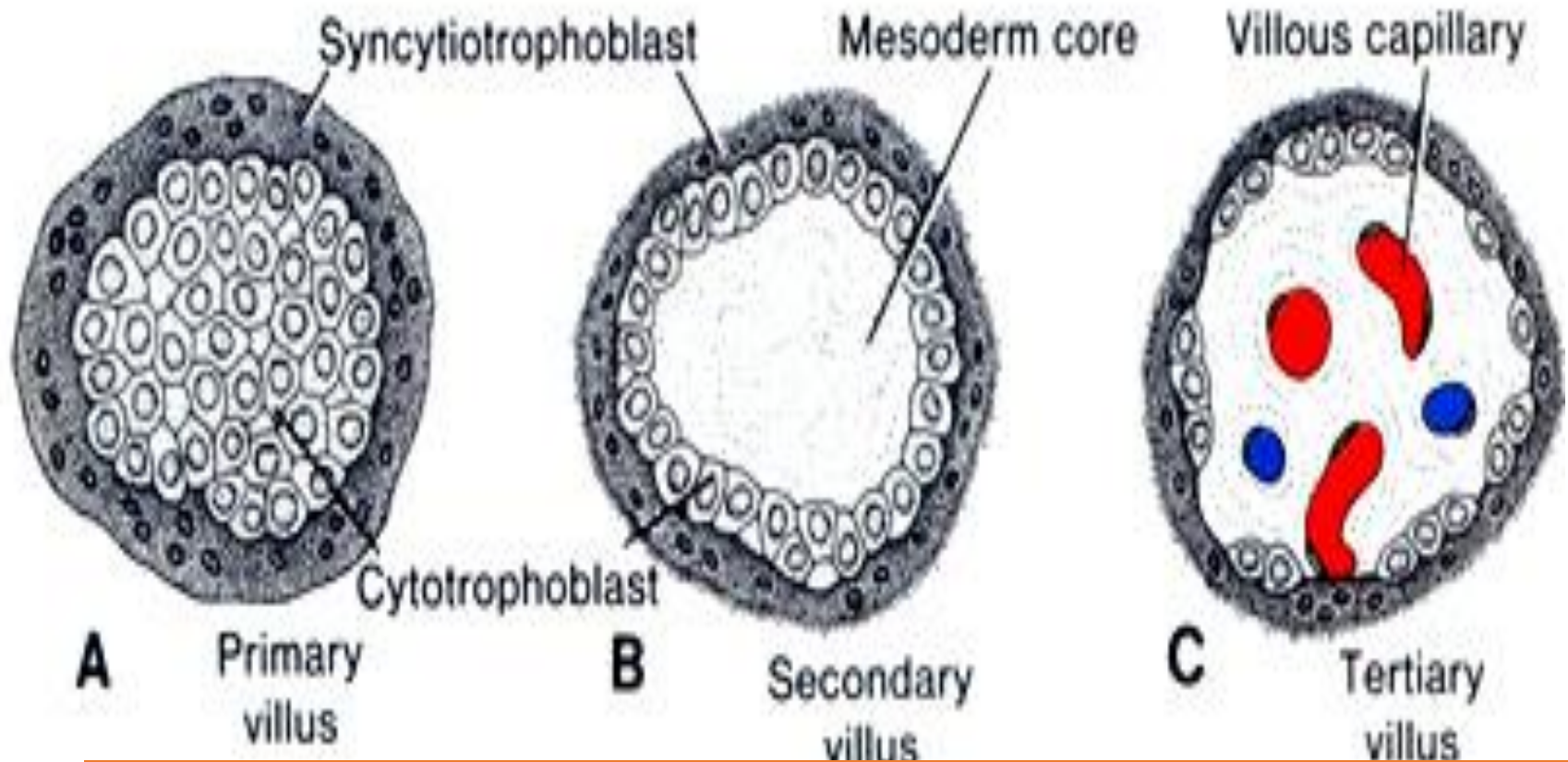
- chorionic villi are non-vascular
- do not have blood vessels for the blood exchange to take place between mother and embryo
- are formed exclusively of trophoblast

- **In the secondary stage**

- chorionic villi become larger, with more ramifications
- the mesoderm starts growing into them
- at this point, they are constituted by trophoblast and mesoderm.

- **In the tertiary stage**

- chorionic villi become vascularized because blood vessels begin growing into the mesoderm
- chorionic villi are thus at this stage made of trophoblast, mesoderm and umbilical arteries and veins (fetal blood vessels).



primary stem villi → secondary stem villi → tertiary stem villi
(with free villi later)

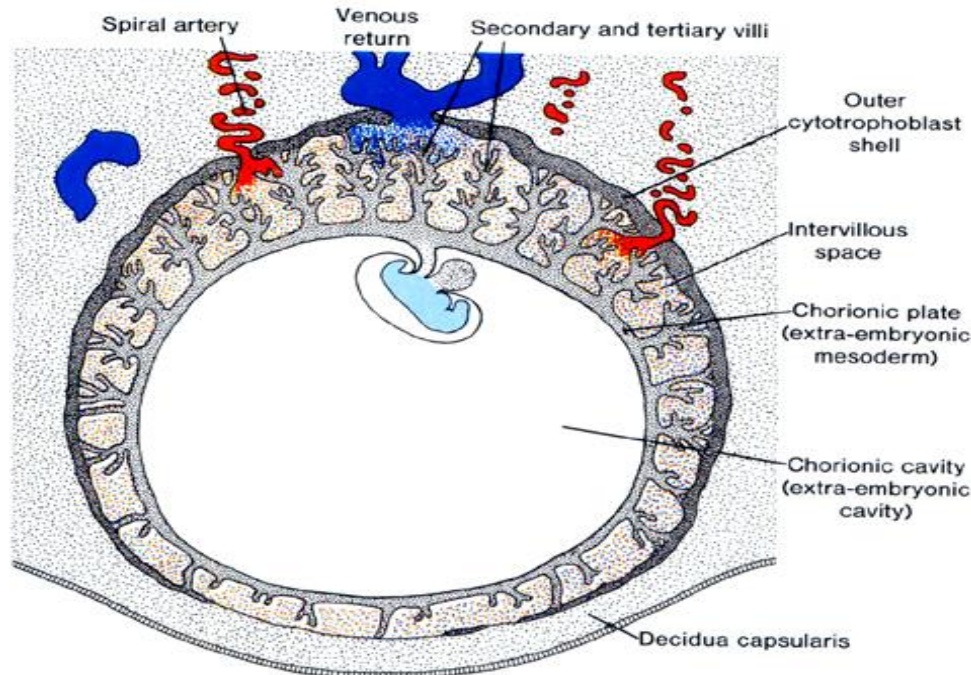
Classification

- **Chorion frondosum**

- ✓ adjacent to decidua basalis with numerous
- ✓ large and branched villi
- ✓ form fetal component of the **placenta**.

- **Chorion leave**

- ✓ associated with decidua capsularis
- ✓ villi of which become degenerated



Functions of Chorion

the chorion has two main functions:
protect the embryo and nurture the embryo.

- To protect the embryo, the chorion produces a fluid known as chorionic fluid
- The chorionic fluid lies in the chorionic cavity, which is the space between the chorion and the amnion
- The chorionic fluid protects the embryo by absorbing shock originating from forces such as movement

Abnormalities

1. hydatidiform mole

(Gestational Trophoblastic Disease; Molar Pregnancy)

- excessive proliferation of trophoblast, edema of stroma → vesicular villi with dead embryo
- is a group of disorders that develop from cells (called trophoblasts) that surround a developing embryo and that eventually form the placenta and amniotic sac
- Gestational trophoblastic disease includes
 - **Tumors that do not invade other tissues:**
 - These tumors include hydatidiform mole, exaggerated placental site, and placental site nodule.
 - These tumors are usually not cancerous, but they may develop into invasive tumors.
 - **Tumors that invade other tissues:** These tumors (called gestational trophoblastic neoplasms) include placental-site trophoblastic tumor, epithelioid trophoblastic tumor, choriocarcinoma, and invasive mole. These tumors are usually cancerous.
- **About 80% of hydatidiform moles are not cancerous.**

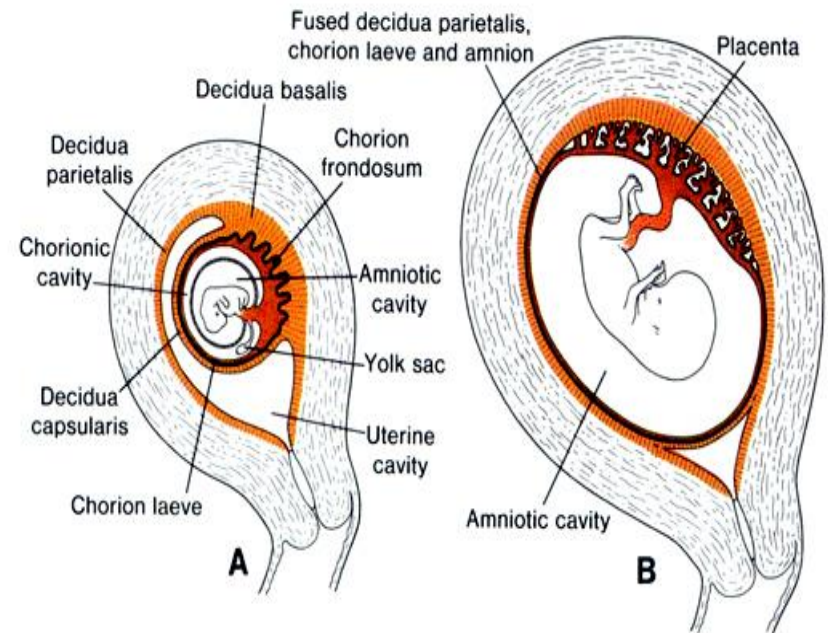
2. chorion carcinoma:

- malignant proliferation of trophoblast cells



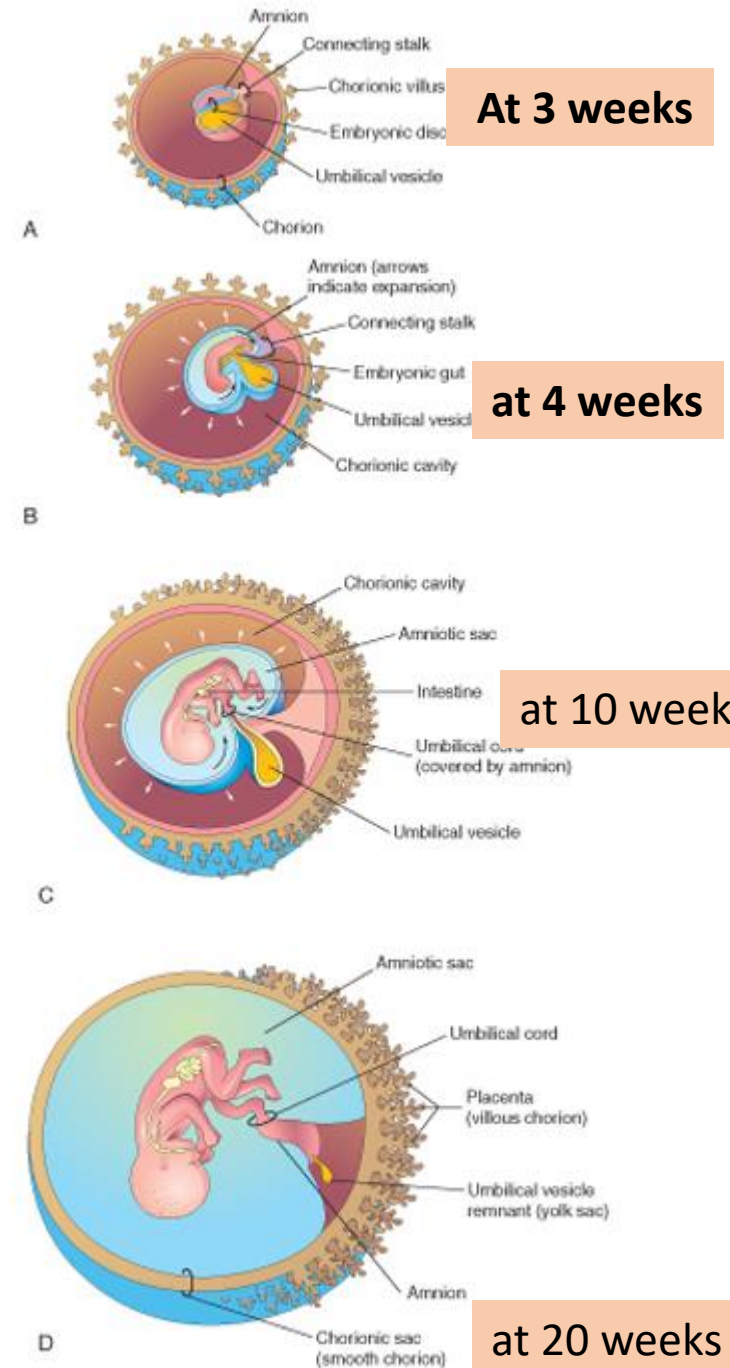
2. Amnion

- thin, transparent membrane which form amniotic cell layer and outer extraembryonic mesoderm layer
- membranous that surrounds the embryo and fetus
- refers to a membranous structure which covers and protects the embryo
- amnion → amniotic cavity → fuses with the chorion → surface of umbilical cord and fetal surface of placenta



- As the amnion enlarges, it gradually **obliterates the chorionic cavity** and forms the epithelial covering of the umbilical cord

Figure 7-22 Illustrations showing how the amnion enlarges, obliterates the chorionic cavity, and envelops the umbilical cord. Observe that part of the umbilical vesicle is incorporated into the embryo as the primordial gut. Formation of the fetal part of the placenta and degeneration of chorionic villi are also shown.;A, B, C, D,.



Amniotic Fluid

- plays a major role in fetal growth and development
- some amniotic fluid is secreted by:
 - **amniotic cells**
 - most is **derived from maternal tissue**, and **interstitial fluid** by diffusion across the **amniochorionic membrane from the decidua parietalis**
 - Later there is diffusion of fluid through the chorionic plate from blood in the intervillous space of the placenta
 - Before keratinization of the skin occurs, a major pathway for passage of water and solutes in tissue fluid from the fetus to the amniotic cavity is through the skin; thus, amniotic fluid is similar to fetal tissue fluid

Fluid is also secreted by the fetal respiratory and gastrointestinal tracts and enters the amniotic cavity

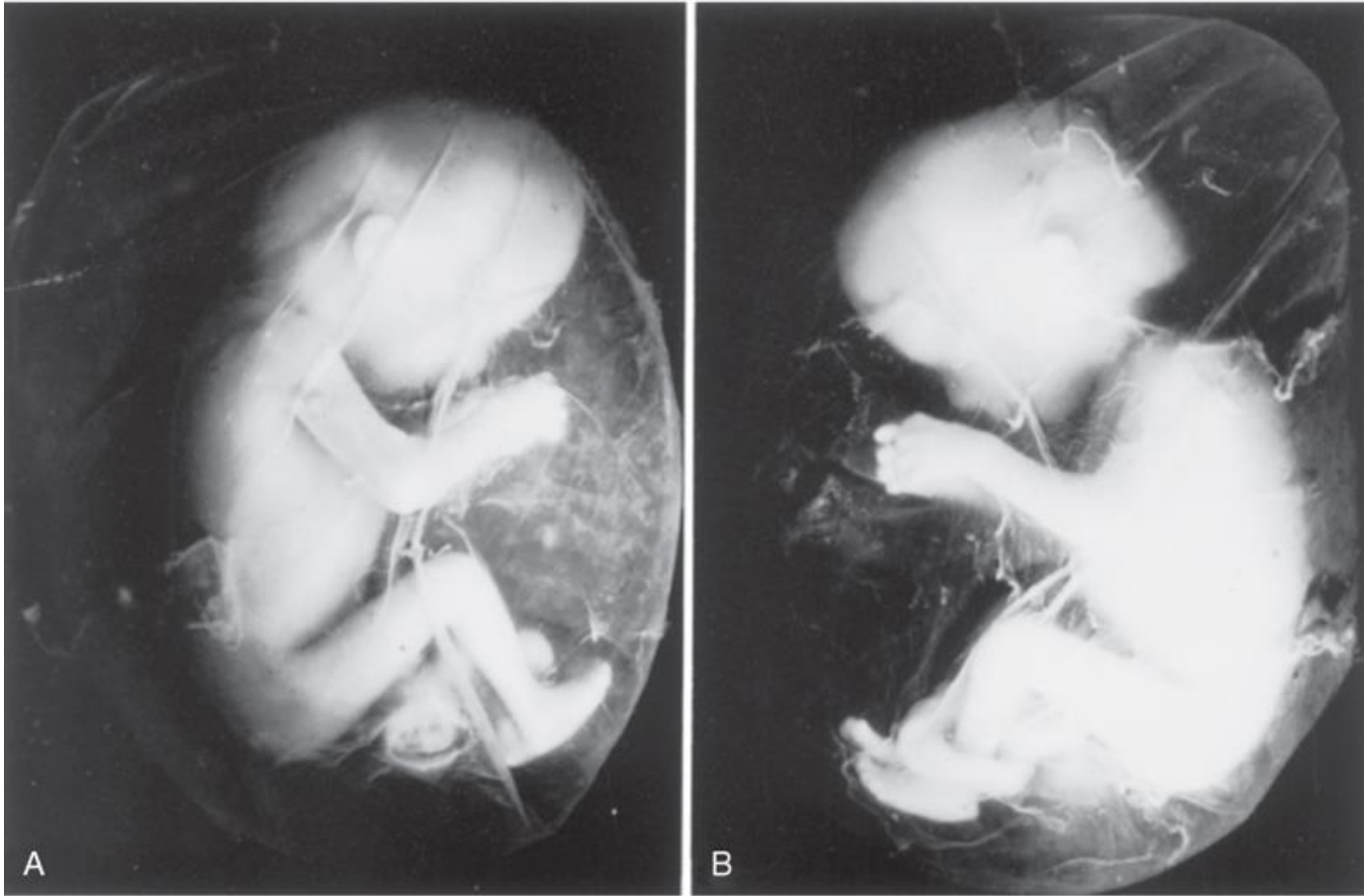
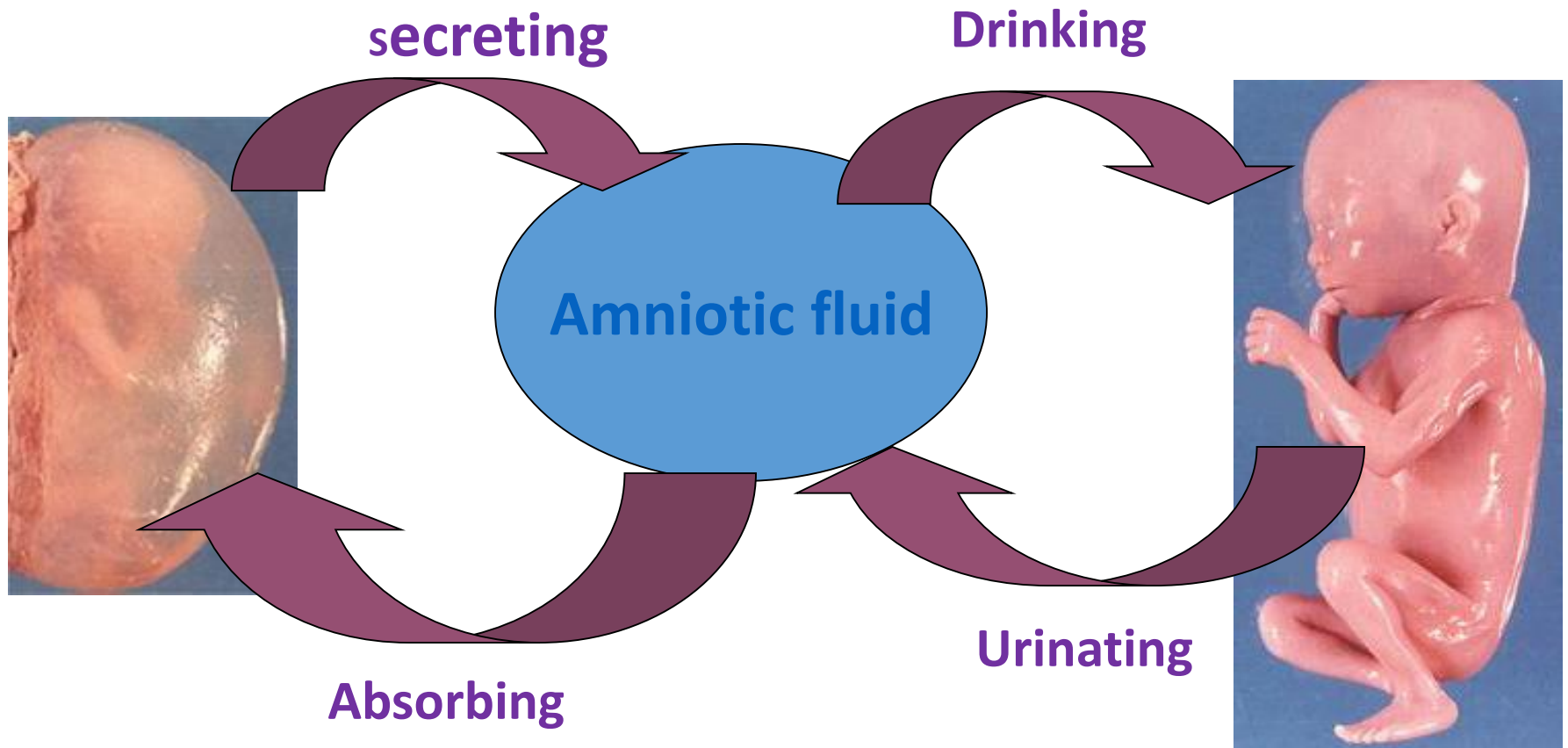


Figure 7-21 **A**, 12-week fetus within its amniotic sac. The fetus and its membranes aborted spontaneously. It was removed from its chorionic sac with its amniotic sac intact. Actual size. **B**, Note that the umbilical cord is looped around the left ankle of the fetus. Coiling of the cord around parts of the fetus affects their development when the coils are so tight that the circulation to the parts is affected.

- **The daily rate of contribution of fluid**
 - to the amniotic cavity from the respiratory tract is **300 to 400 mL**
 - Beginning in the **11th week**
 - the fetus contributes to the amniotic fluid by excreting urine into the amniotic cavity
 - **By late pregnancy**, approximately **500 mL of urine** is added daily.
 - The volume of amniotic fluid normally increases slowly reaching
 - ✓ **≈ 30 mL at 10 weeks**
 - ✓ **350 mL at 20 weeks**
 - ✓ and **700 to 1000 mL by 37 weeks**
- water content of the fluid is totally renewed in every 3h

Circulation of amniotic fluid



Amniotic fluid

- **is swallowed by the fetus and absorbed by the fetus's respiratory and digestive tracts**
- during the final stages of pregnancy
 - **the fetus swallows up to 400 mL of amniotic fluid per day**
- The fluid passes into the fetal bloodstream and the waste products in it cross the placental membrane
- and enter the maternal blood in the intervillous space.
- Excess water in the fetal blood is excreted by the fetal kidneys and returned to the amniotic sac through the fetal urinary tract

Composition of Amniotic Fluid

- Composed of an aqueous solution in which undissolved material (desquamated fetal epithelial cells) is suspended
- **contains approximately**
 - equal portions of organic and inorganic salts.
 - half of the organic constituents are protein
 - the other half consists of carbohydrates, fats, enzymes, hormones, and pigments
- **As pregnancy advances, the composition of the amniotic fluid changes as fetal excreta (meconium [fetal feces] and urine) are added.**

- **Because fetal urine enters the amniotic fluid,**
 - studies of **fetal enzyme systems, amino acids, hormones, and other substances** can be conducted on fluid removed by **amniocentesis**
 - Studies of cells in the amniotic fluid permit diagnosis of chromosomal abnormalities such as trisomy **21 (Down syndrome)**
 - **Moreover, fluorescent in situ hybridization** analyses are often carried out on fetal cells if there is a family history or clinical indication of certain **genetic disorders**.
 - **High levels of alpha fetoprotein** usually indicate the presence of a severe neural tube defect
 - **Low levels of alpha fetoprotein** may indicate chromosomal aberrations such as **trisomy 21**

Function of amnionic fluid

- Provides a protective cushion and absorb jolts
- Helps to dilate and clean the cervical canal making childbirth easier
- Permits symmetric external growth of the embryo and fetus
- Acts as a barrier to infection
- Permits normal fetal lung development
- Cushions the embryo and fetus against injuries by distributing impacts the mother receives
- Helps control the embryo's body temperature by maintaining a relatively constant temperature
- Assists in maintaining homeostasis of fluid and electrolytes Prenatal diagnosis: biochemical assay (AFP), chromosome or gene analysis.

Abnormalities

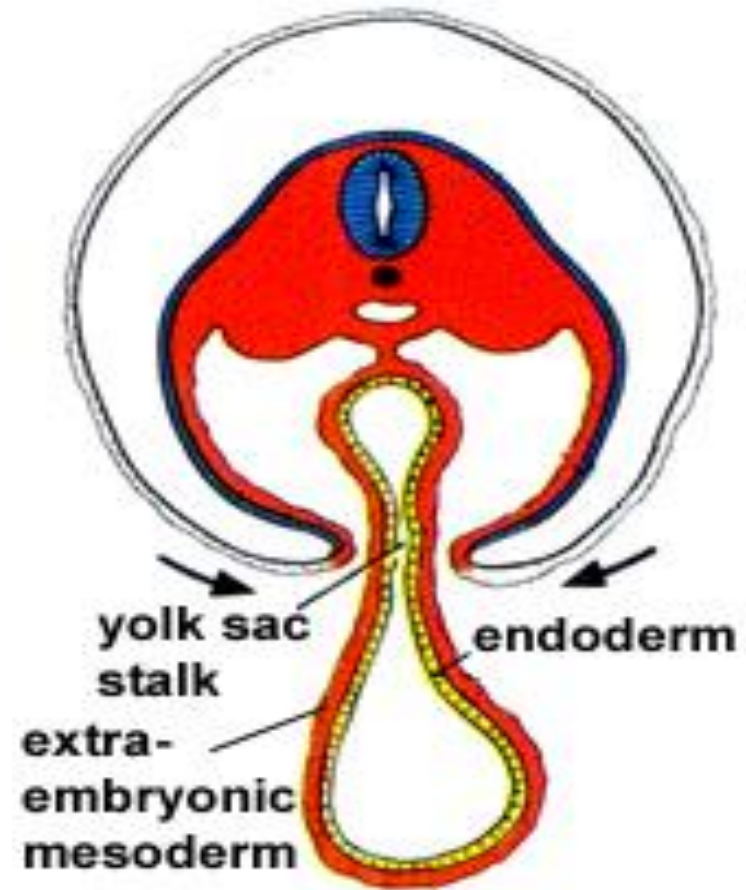
- **polyhydramnios**, amniotic fluid >2 liter, suggesting anencephaly, hydrocephalus, or esophageal atresia
- **oligohydramnios**, amniotic fluid <500 ml, due to renal agenesis or urethral atresia
- **Rupture of the amniochorionic membrane** is the most common event leading to premature labor and delivery and the most common complication resulting in oligohydramnios
- **Rupture of the amnion** may cause various fetal anomalies that constitute the **amniotic band syndrome**

3. The umbilical vesicle (yolk sac)

- The umbilical vesicle (yolk sac) can be observed **sonographically early in the 5th week**
- At 32 days, the umbilical vesicle is large
- By 10 weeks, the umbilical vesicle has shrunk to a pear-shaped remnant approximately **5 mm in diameter**
- and is connected to the midgut by a narrow omphaloenteric duct (yolk stalk).
- By 20 weeks, the umbilical vesicle is very small thereafter, it is usually not visible

Yolk sac

- 1) Inner **endoderm** + outer **extraembryonic**
- **mesoderm**

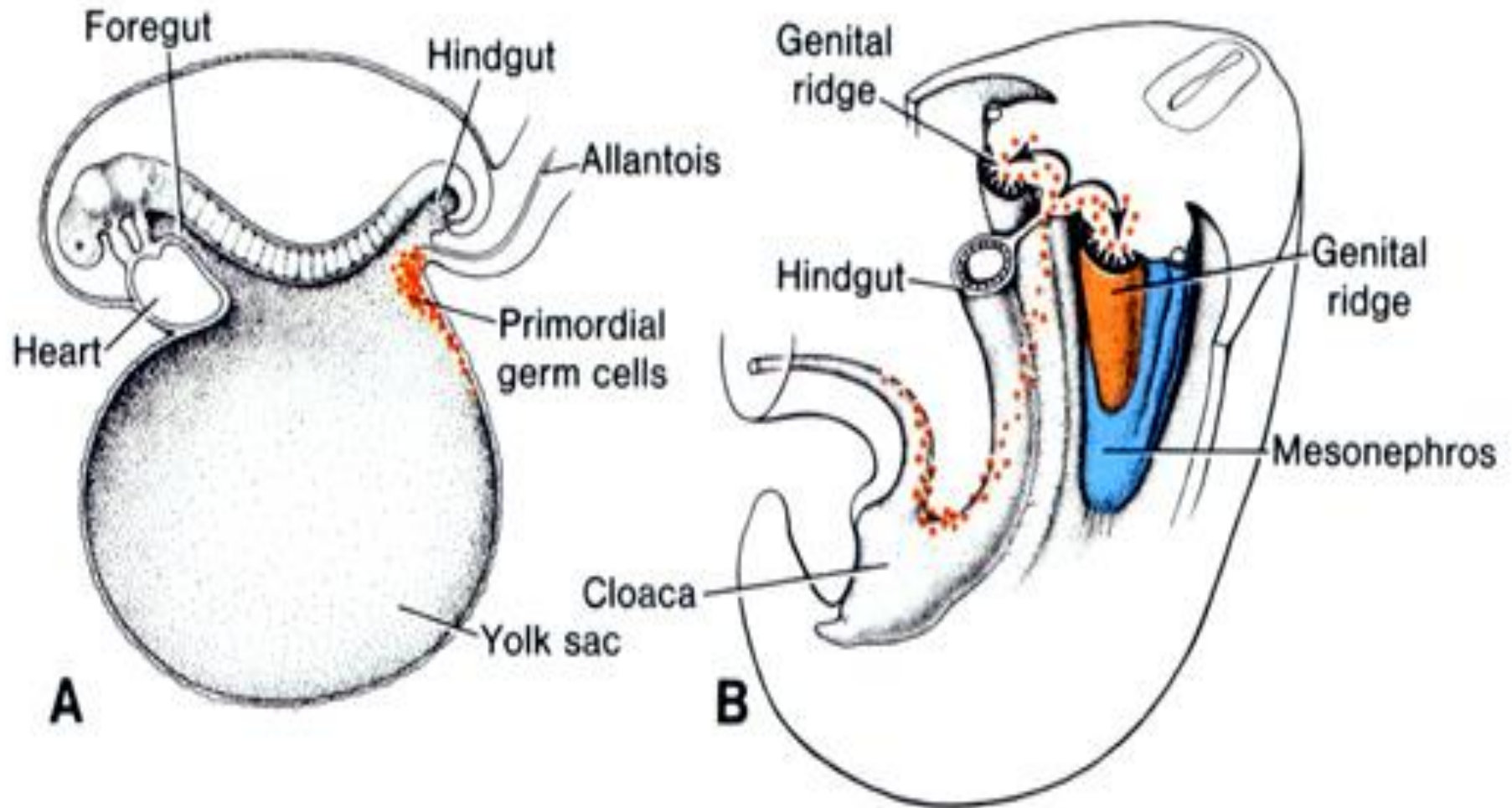


- **Significance of the Umbilical Vesicle**

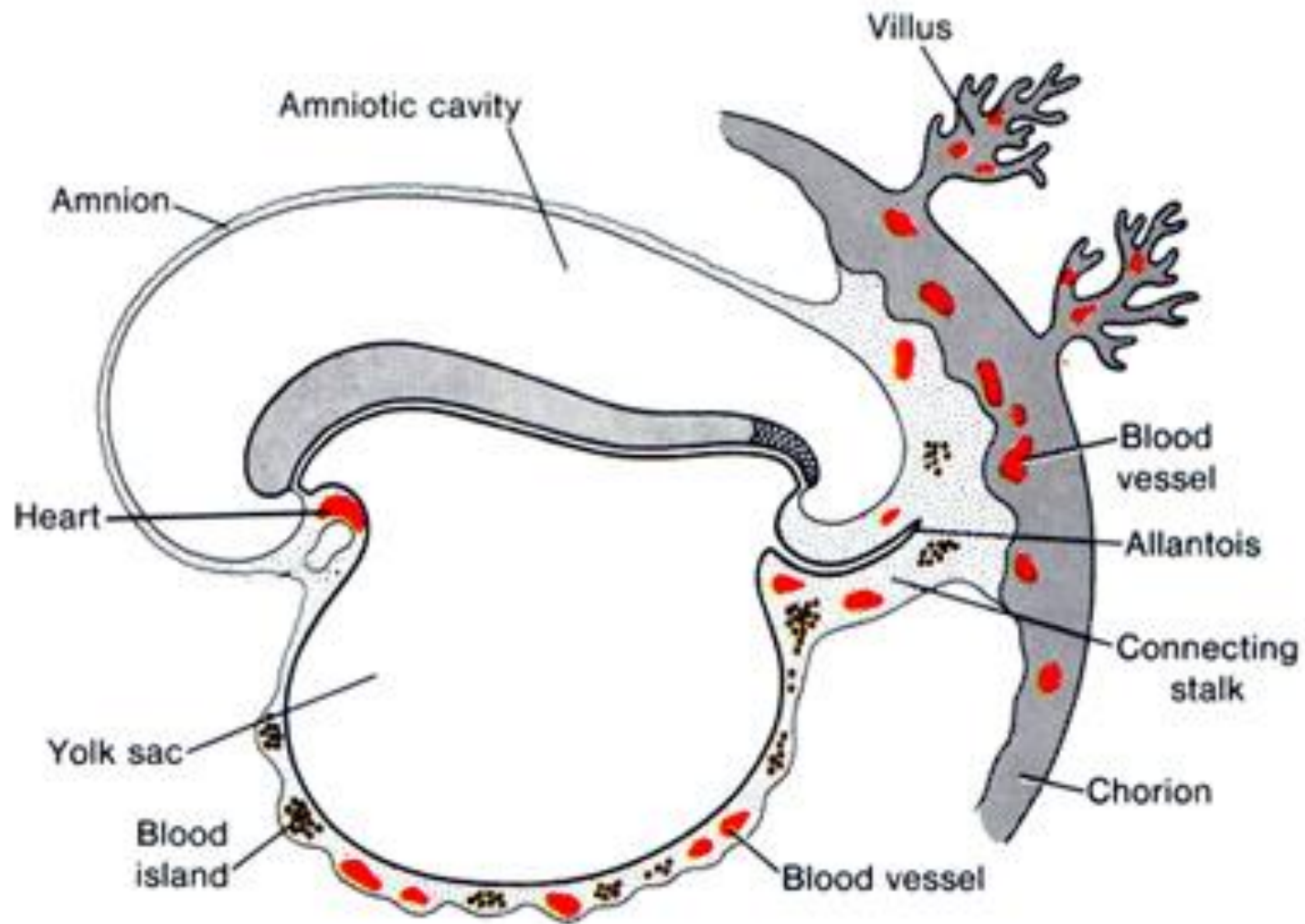
- It has a role in the transfer of nutrients to the embryo during the second and third weeks when the uteroplacental circulation is being established
- **Blood development first occurs** in the well-vascularized extraembryonic mesoderm covering the wall of the umbilical vesicle beginning in the third week and continues to form there until **hemopoietic activity begins in the liver during the sixth week**
- During the **4th week**, the endoderm of the umbilical vesicle is incorporated into the embryo as the primordial gut
- **Its endoderm**, derived from epiblast, gives rise to the epithelium of the trachea, bronchi, lungs, and digestive tract
- Primordial germ cells appear in the endodermal lining of the wall of the umbilical vesicle in the third week
- and subsequently migrate to the developing

Significance in development:

- endoderm cells → **primordial germ cells**



- mesoderm cells → **primitive blood cells**



4. The allantois

- it appears as a sausage-like diverticulum from the caudal wall of the umbilical vesicle that extends into the connecting stalk
- **During the 2nd month, the extraembryonic part of the allantois degenerates**

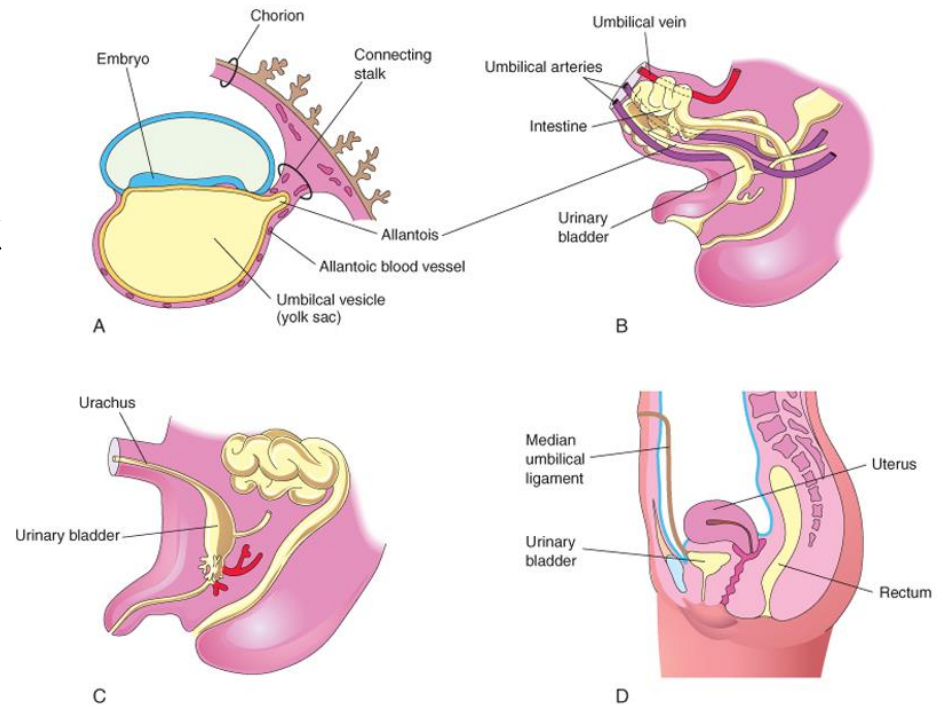


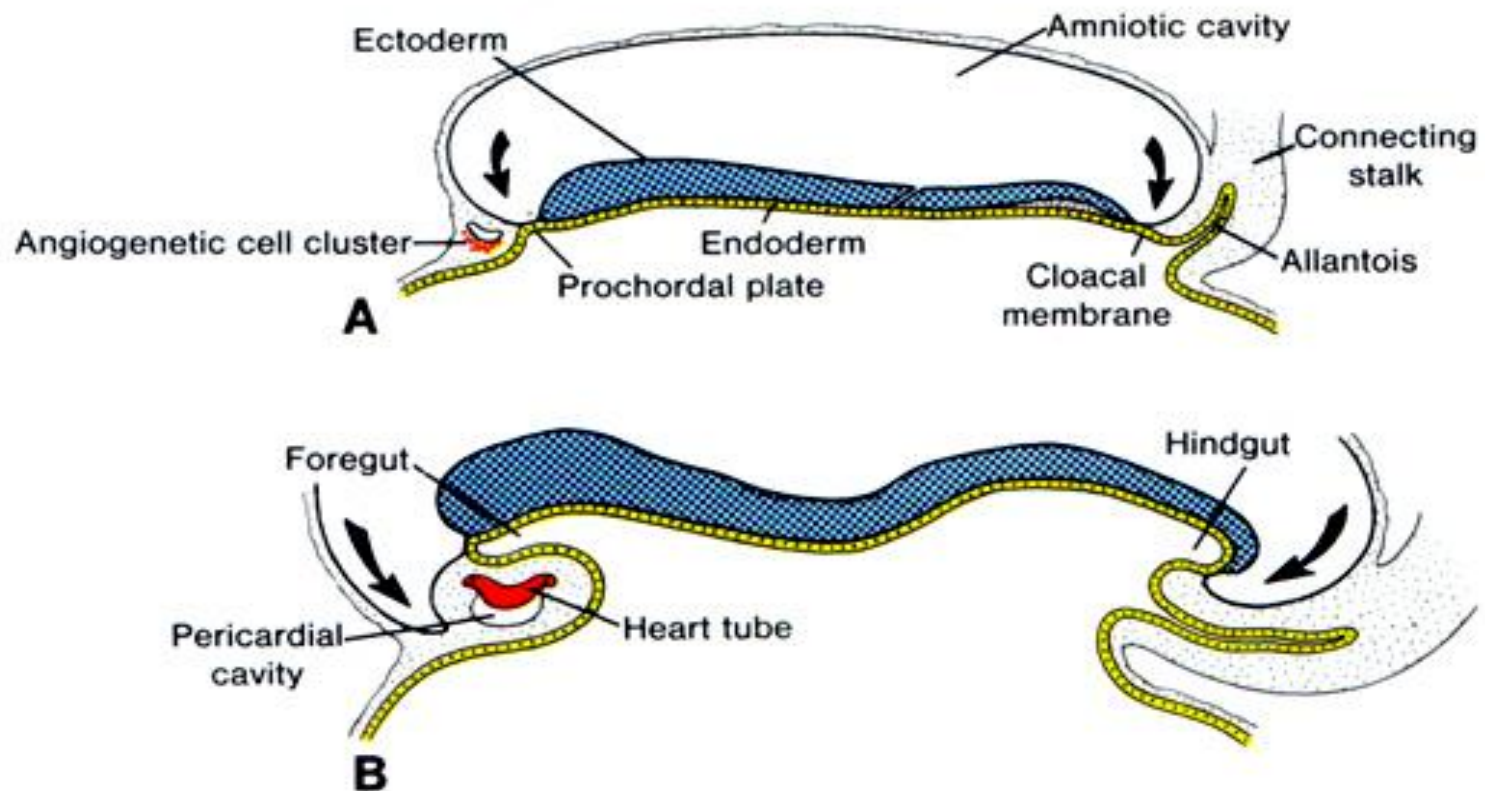
Figure 7-24 Illustrations of the development and usual fate of the allantois. **A**, A 3-week embryo. **B**, A 9-week fetus. **C**, A 3-month male fetus. **D**, Adult female. The nonfunctional allantois forms **the urachus** in the fetus and the median umbilical ligament in the adult.

Importance of the Allantois

- Although the allantois is not functional in human embryos, it is important for three reasons:
 - **Blood formation** occurs in its wall during the third to fifth weeks.
 - Its blood vessels persist as the umbilical vein and arteries
 - The intraembryonic part of the allantois runs from the umbilicus to the urinary bladder, with which it is continuous
 - As the bladder enlarges, the allantois involutes to form a thick **tube, the urachus**
 - After birth, **the urachus** becomes **a fibrous cord**, the median umbilical ligament, that extends from the apex of the urinary bladder to the umbilicus.

Allantois

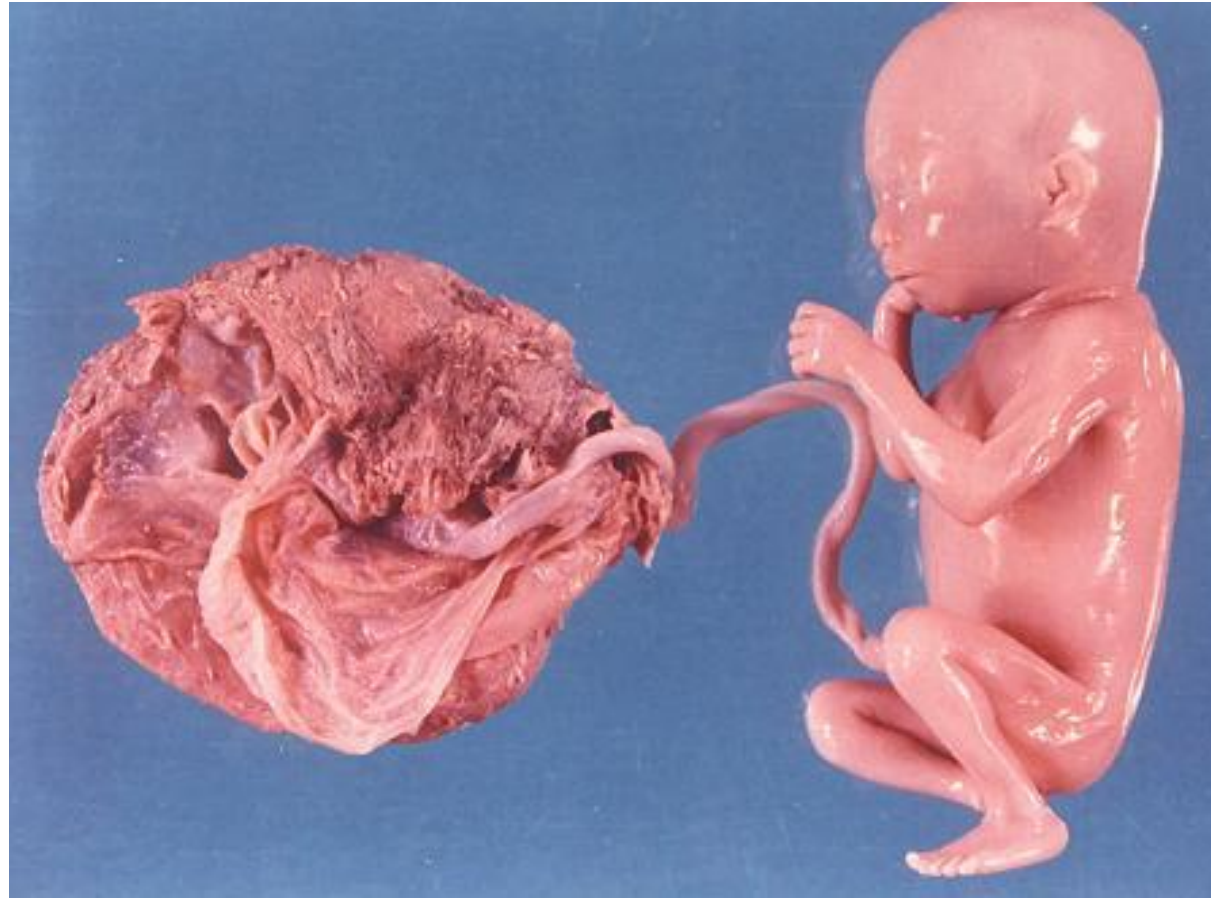
- Formation: caudal part of yolk sac → diverticulum extending into body stalk
- significance: blood vessels in its wall → umbilical arteries and vein



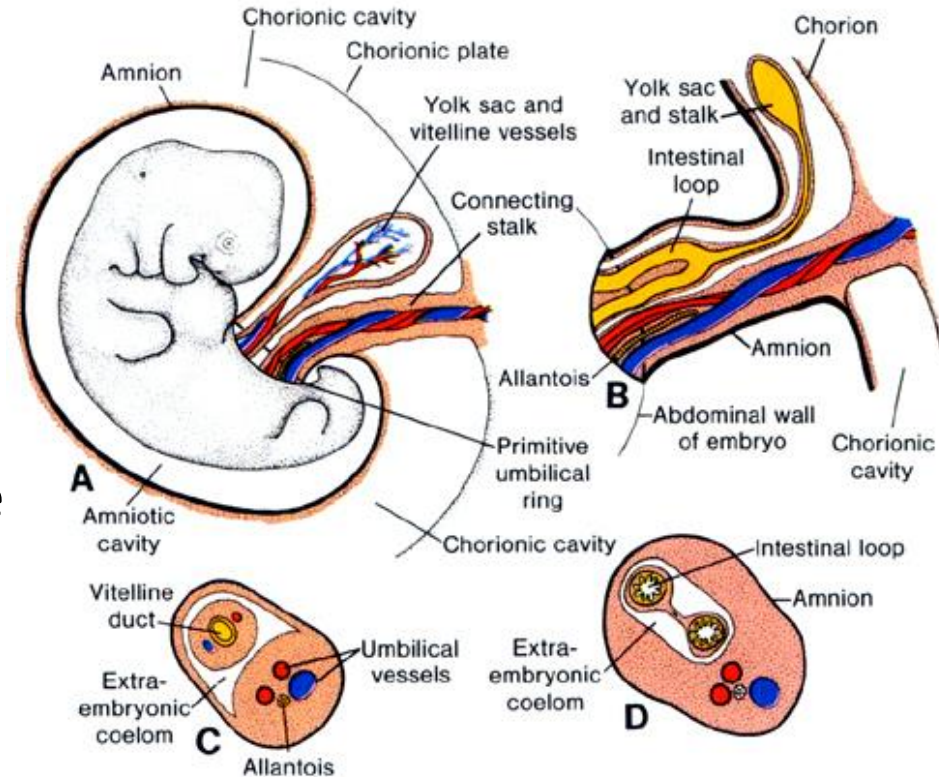
5. The Umbilical Cord

- Formation:
 - from the body stalk → connecting the embryo or fetus to the placenta, **50 cm long** at birth
 - The attachment of the umbilical cord to the placenta is usually near the center of the fetal surface of this organ

- it may attach at any point
- For example, insertion of it at the placental margin produces a **battledore placenta**
- its attachment to the fetal membranes is a velamentous insertion of the cord



- Contents: body stalk, yolk sac stalk, allantois,
- umbilical vessels → **2 arteries and 1 vein, mucoid connective tissue**
- umbilical cord is usually **1 to 2 cm** in diameter and **30 to 90 cm in** length (average, 55 cm)
- Excessively long or short cords are uncommon
- Long cords have a tendency to prolapse and/or to coil around the fetus



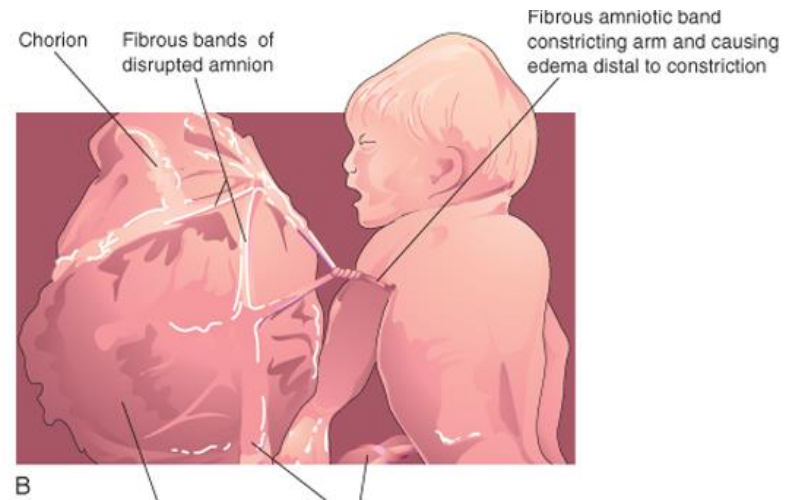
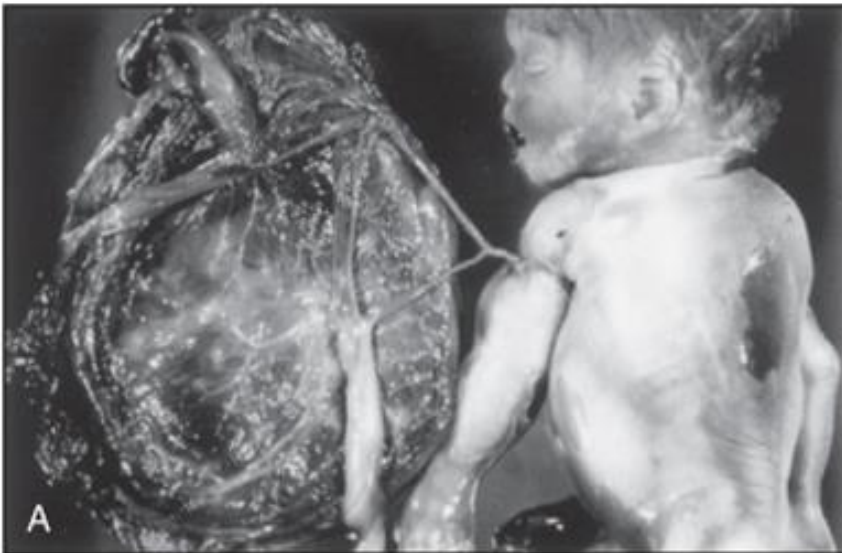
Abnormalities

very long ($> 90\text{cm}$)

- encircling neck or extremity

- very short ($< 30\text{cm}$)

- prematurely detaching of placenta and bleeding



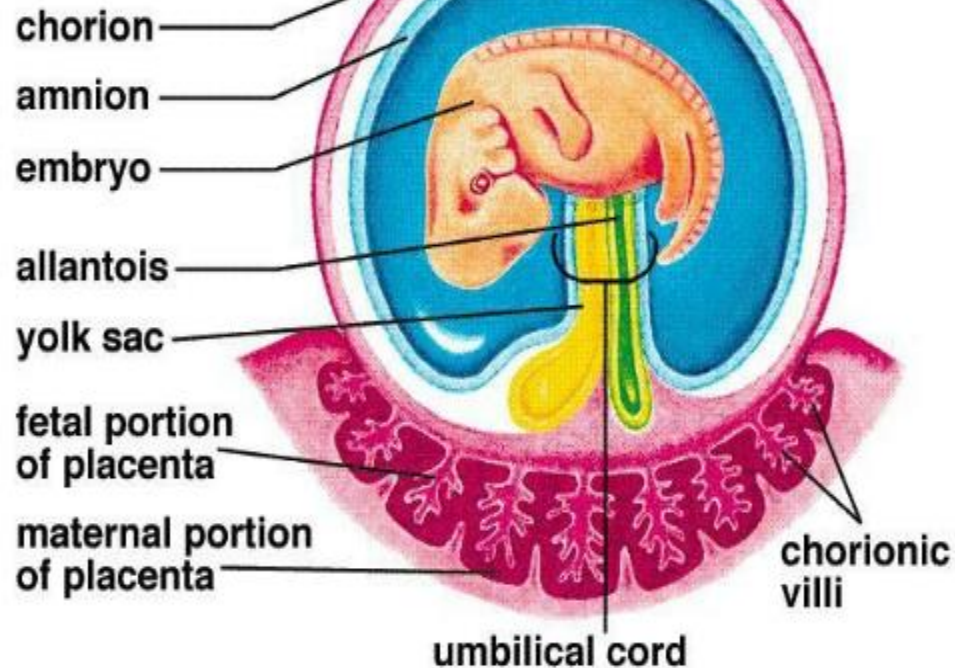
2. The placenta

- separate the fetus from the **endometrium**
 - Serve as medium of interchanging of substances such as:
 - **nutrients and oxygen and waste substances**
 - **occurs between the maternal and fetal bloodstreams through the placenta**
- The vessels in the umbilical cord:
 - **connect the placental circulation with the fetal circulation**

THE PLACENTA

- is the primary site of nutrient and gas exchange between **the mother and fetus**
- is a fetomaternal organ that has two components:
 - a. **A fetal part that develops from the chorionic sac**
 - b. **A maternal part that is derived from the endometrium**

Extraembryonic Membranes



- **The placenta and umbilical cord form:**
 - a transport system for substances passing between the mother and fetus
 - Nutrients and oxygen pass from the maternal blood through the placenta to the fetal blood
 - waste materials and carbon dioxide pass from the fetal blood through the placenta to the maternal blood
- The placenta and fetal membranes perform the following functions and activities:
 - **Protection**
 - **Nutrition**
 - **Respiration**
 - **Excretion**
 - **and hormone production.**

***Shortly after birth, the placenta and fetal membranes are expelled from the uterus as the afterbirth.

The Decidua

- refers to the gravid endometrium,
- Is the functional layer of the endometrium in a pregnant woman
- separates from the rest of the uterus after parturition (childbirth)

- The three regions of the decidua and are named according to their relation to **the implantation site**

1. The decidua basalis

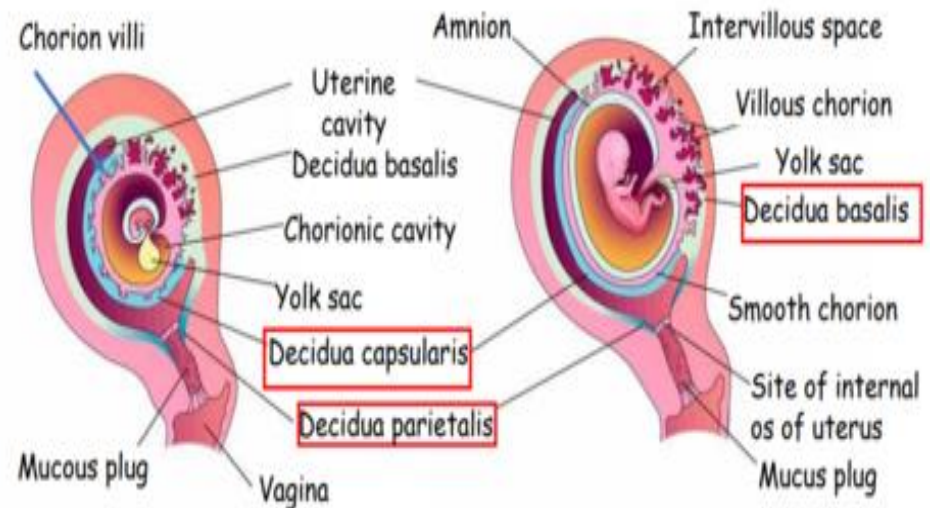
- ✓ is the part of the decidua deep to the conceptus that forms the maternal part of the placenta.

2. The decidua capsularis

- ✓ is the superficial part of the decidua overlying the conceptus.

3. The decidua parietalis

- ✓ is all the remaining parts of the decidua.



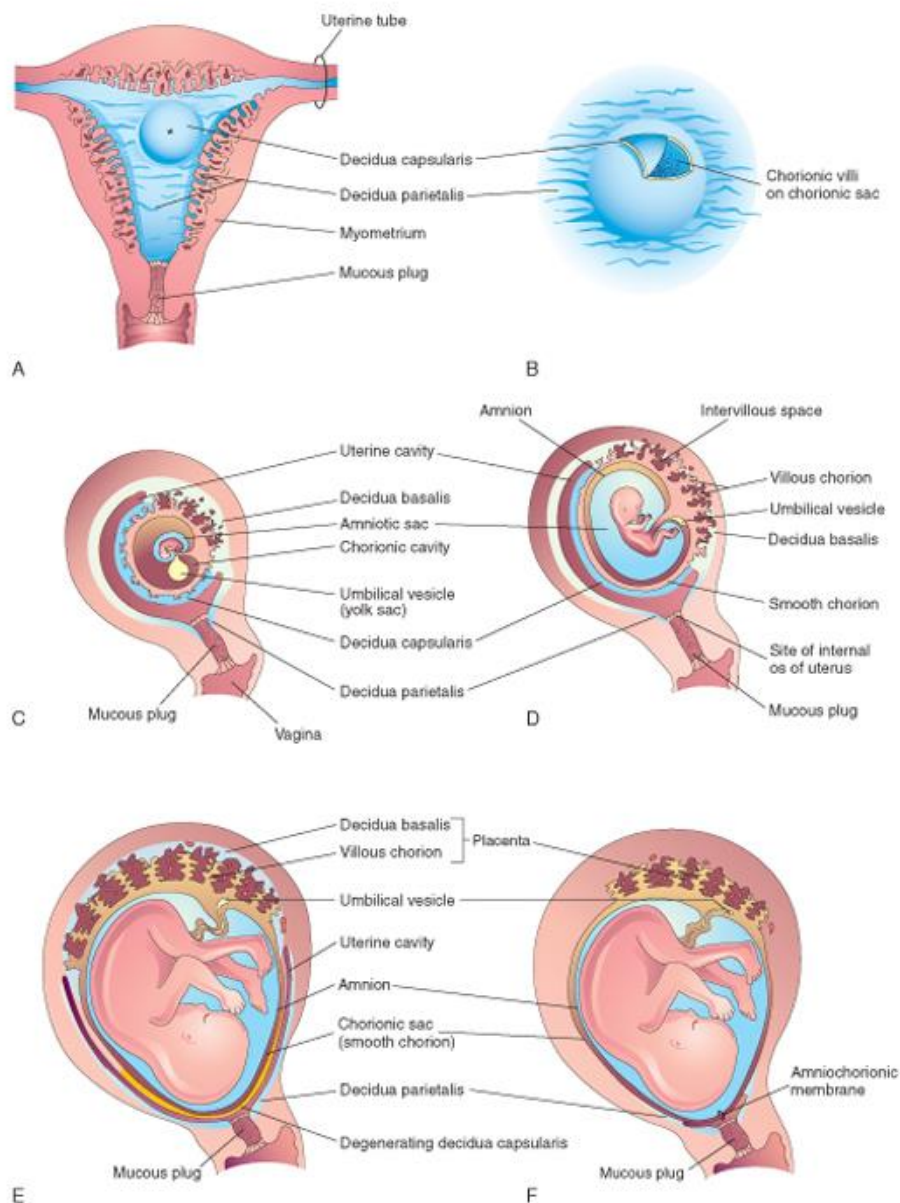


Figure 7-1 Development of the placenta and fetal membranes. **A**, Frontal section of the uterus showing elevation of the decidua capsularis by the expanding chorionic sac of a 4-week embryo implanted in the endometrium on the posterior wall (*). **B**, Enlarged drawing of the implantation site. The chorionic villi were exposed by cutting an opening in the decidua capsularis. **C** to **F**, Sagittal sections of the gravid uterus from weeks 5 to 22 showing the changing relations of the fetal membranes to the decidua. In **F**, the amnion and chorion are fused with each other and the decidua parietalis, thereby obliterating the uterine cavity. Note in **D** to **F** that the chorionic villi persist only where the chorion is associated with the decidua basalis.

- In response to increasing progesterone levels in the maternal blood, the connective tissue cells of the decidua enlarge to form **pale-staining decidual cells**
- These cells enlarge as **glycogen and lipid accumulate in their cytoplasm**
- The cellular and vascular changes occurring in the endometrium as the blastocyst implants **constitute the decidual reaction**
- **Many decidual cells degenerate:**
 - near the chorionic sac in the region of the syncytiotrophoblast
 - maternal blood
 - uterine secretions,

provide a rich source of nutrition for the embryo
- **The full significance of decidual cells:**
 - is not understood
 - but it has also been suggested that they protect the maternal tissue against uncontrolled invasion by the syncytiotrophoblast
 - and that they may be involved in hormone production

Decidual regions, clearly recognizable during ultrasonography, are important in diagnosing early pregnancy.

Development of the Placenta

- Early placental development is characterized by
 - the rapid proliferation of the trophoblast
 - development of the chorionic sac and chorionic villi
 - **By the end of the 3rd week**, the anatomic arrangements necessary for physiologic exchanges between the mother and her embryo are established.
 - A complex vascular network is established in the placenta by the end of the 4th week,
 - which facilitates **maternal-embryonic exchanges** of gases, nutrients, and metabolic waste products.

- **Chorionic villi:**

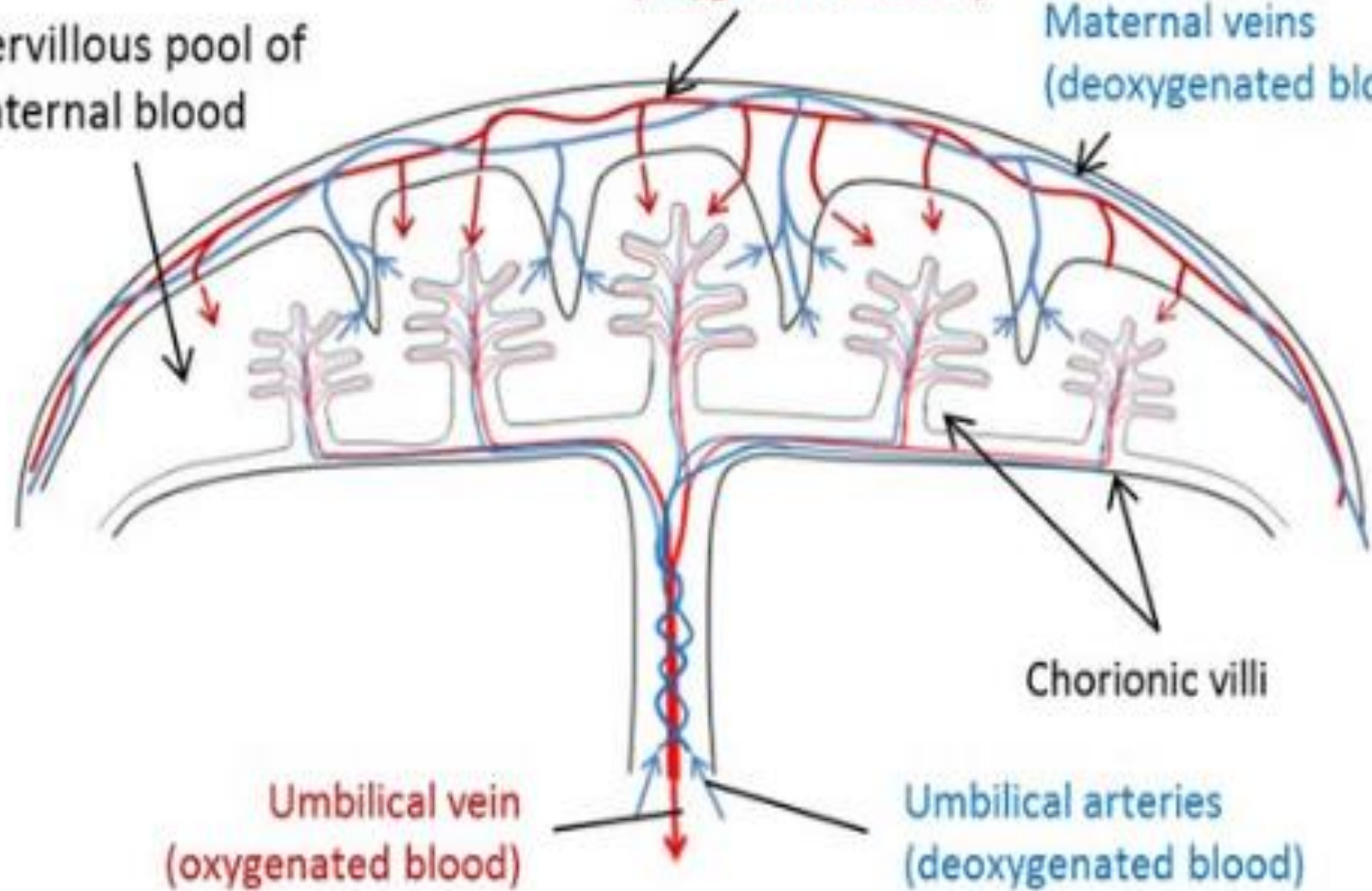
- are **tiny projections of placental tissue that look like fingers,**
- **and contain the same genetic material as the fetus**
- cover the entire chorionic sac (**chorionic frondosum**) until the beginning of the **8th week**
- the villi associated with the decidua capsularis are compressed, reducing the blood supply to them
- These villi soon degenerate, producing a relatively avascular bare area, **the smooth chorion (chorionic Leave).**
- As these villi disappear, those associated with the decidua basalis rapidly increase in number, branch profusely, and enlarge.
- This bushy area of the chorionic sac is the **villous chorion**

human placenta

Intervillous pool of maternal blood

Maternal arteries
(oxygenated blood)

Maternal veins
(deoxygenated blood)



Chorionic villi

Umbilical vein
(oxygenated blood)

Umbilical arteries
(deoxygenated blood)

- **Growth in the size and thickness of the placenta** continues rapidly until the fetus is **≈ 18 weeks old** (20 weeks' gestation)
- The fully developed placenta covers **15% to 30%** of the decidua and weighs approximately one sixth that of the fetus
- The placenta has two parts:

a. The fetal part of the placenta

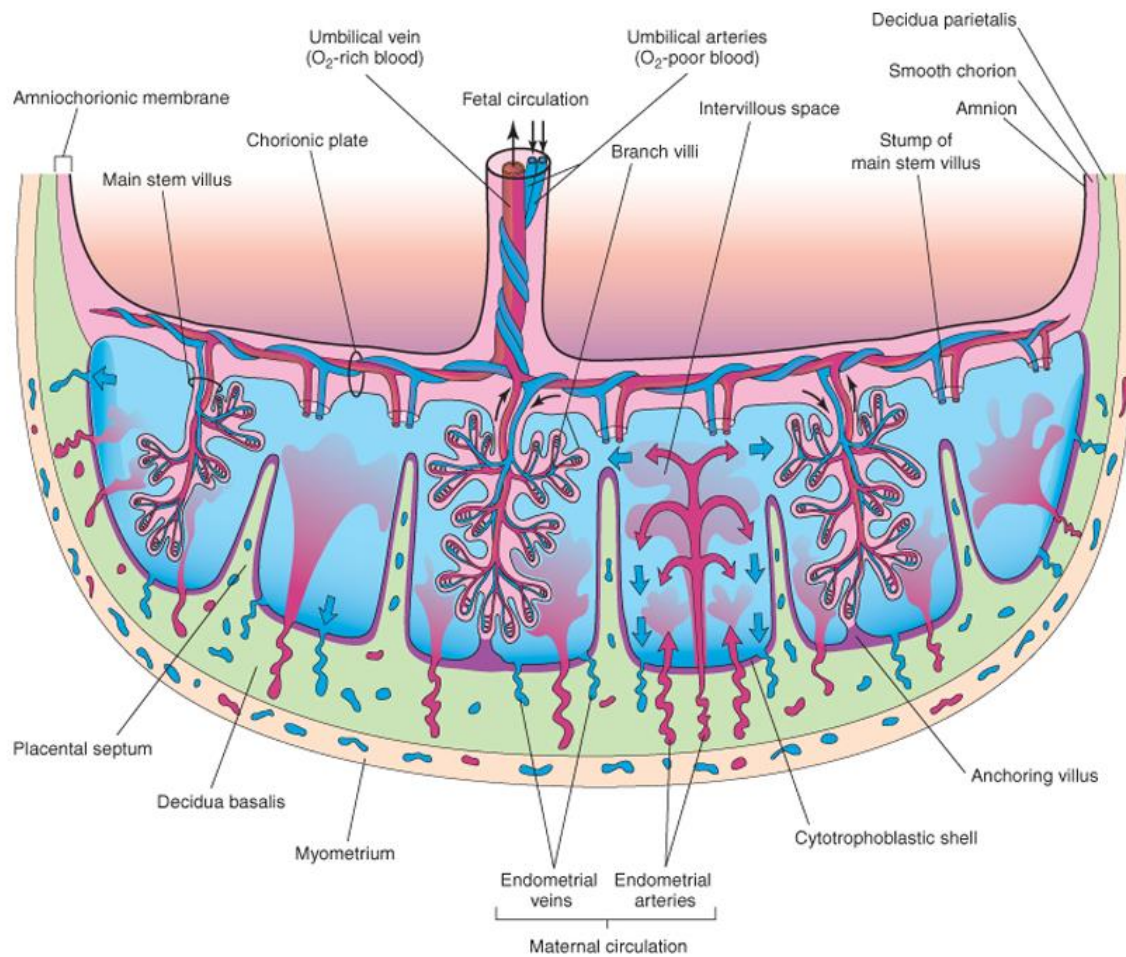
- is formed by the villous chorion
- the chorionic villi that arise from it project into the intervillous space **containing maternal blood**

b. The maternal part of the placenta

- is formed by the decidua basalis, the part of the decidua related to the fetal component of the placenta
- **By the end of the 4th month**, the decidua basalis is almost entirely replaced by the **fetal part of the placenta**

- **The fetal part of the placenta (villous chorion):**

- is attached to the maternal part of the placenta (decidua basalis) by **the cytotrophoblastic shell**
 - ✓ the external layer of trophoblastic cells on the maternal surface of the placenta
- **The chorionic villi:**
 - ✓ attach firmly to the decidua basalis through the **cytotrophoblastic shell**
 - ✓ anchor the chorionic sac to the decidua basalis.
- **Endometrial arteries and veins:**
 - pass freely through gaps in the **cytotrophoblastic shell**,
 - and **open into the intervillous space.**



Schematic drawing of a transverse section through a full-term placenta, showing (1) the relation of the villous chorion (fetal part of placenta) to the decidua basalis (maternal part of placenta), (2) the fetal placental circulation, and (3) the maternal placental circulation. Maternal blood flows into the intervillous space in funnel-shaped spurts from the spiral endometrial arteries, and exchanges occur with the fetal blood as the maternal blood flows around the branch villi. It is through these villi that the main exchange of material between the mother and embryo/fetus occurs.

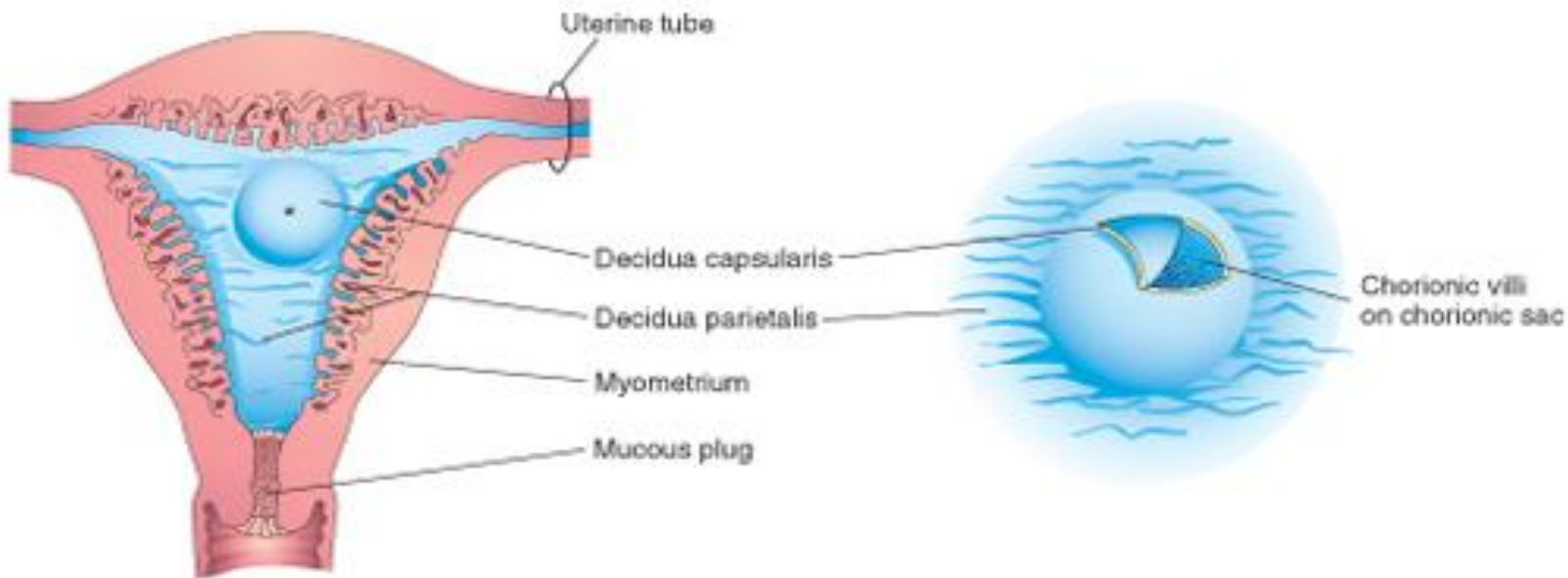
The inflowing arterial blood pushes venous blood out of the intervillous space into the endometrial veins, which are scattered over the surface of the decidua basalis.

Note that the cotyledons are separated from each other by placental septa, projections of the decidua basalis. Each cotyledon consists of two or more main stem villi and many branch villi. In this drawing, only one stem villus is shown in each cotyledon, but the stumps of those that have been removed are indicated.

- **The shape of the placenta**

- is determined by the persistent area of chorionic villi
- Usually a discoid shape.
- **As the chorionic villi invade the decidua basalis,**
 - decidual tissue is eroded to enlarge the intervillous space
 - This erosion produces several wedge-shaped areas of decidua, **placental septa**, that project toward the chorionic plate
 - **The placental septa** divide the fetal part of the placenta into irregular convex areas-**cotyledons**
 - Each cotyledon consists of two or more stem villi and their many branch villi
 - By the **end of the 4th month**,
 - ✓ the decidua basalis is almost entirely replaced by the cotyledons

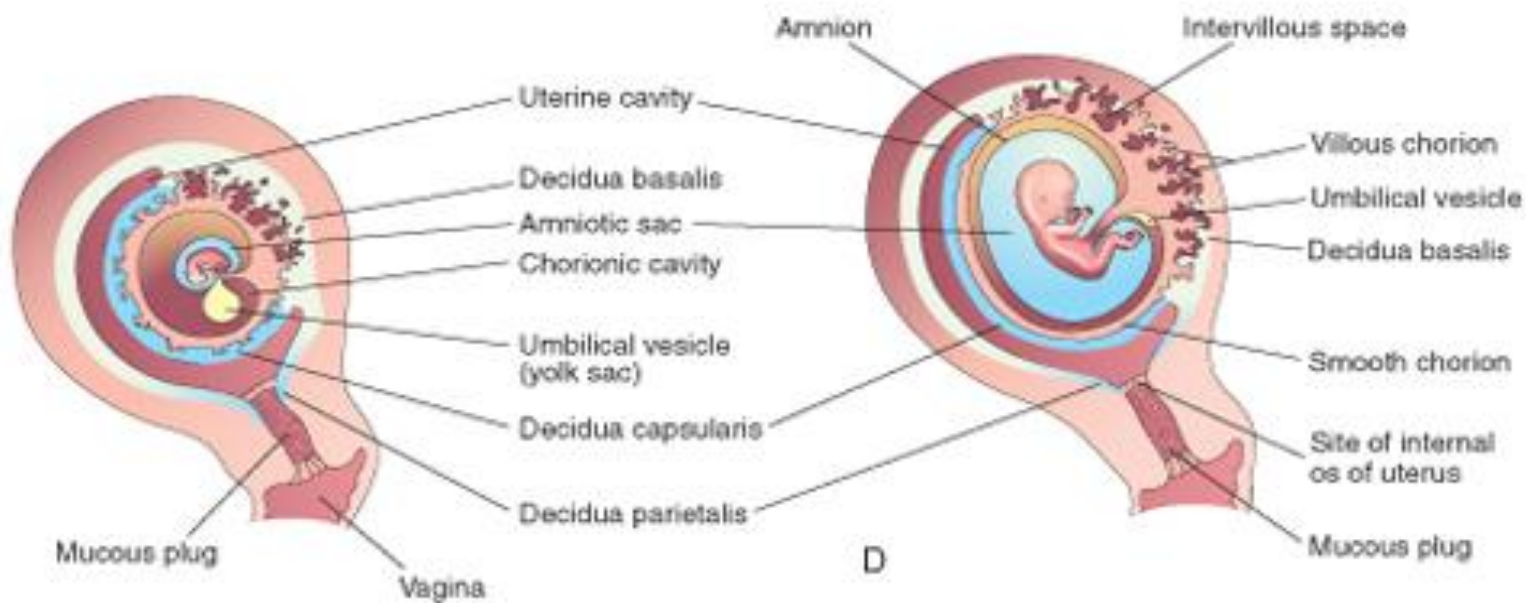
- **The decidua capsularis:**
 - the layer of decidua overlying the implanted chorionic sac
 - forms a capsule over the external surface of the sac
 - As the conceptus enlarges, the decidua capsularis bulges into the uterine cavity and becomes greatly attenuated
 - Eventually the decidua capsularis contacts,
 - and fuses with the decidua parietalis, thereby slowly obliterating the uterine cavity
 - **By 22 to 24 weeks,**
 - the reduced blood supply to the decidua capsularis causes it to degenerate and disappear.



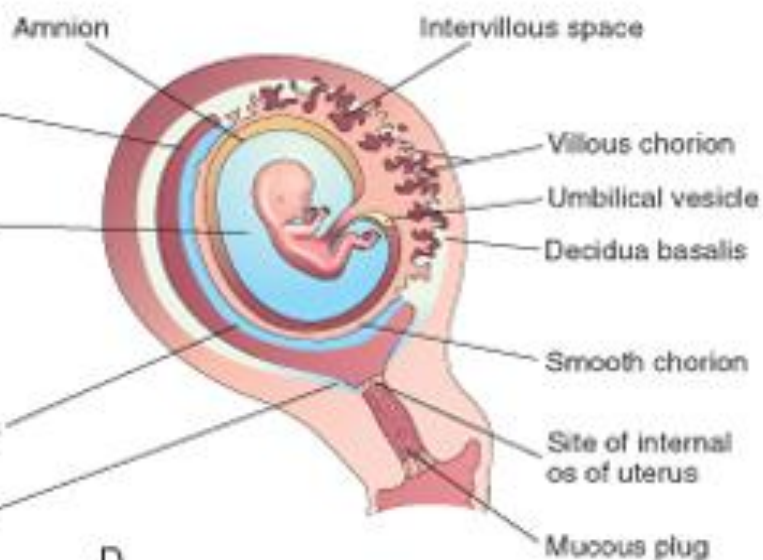
A



B

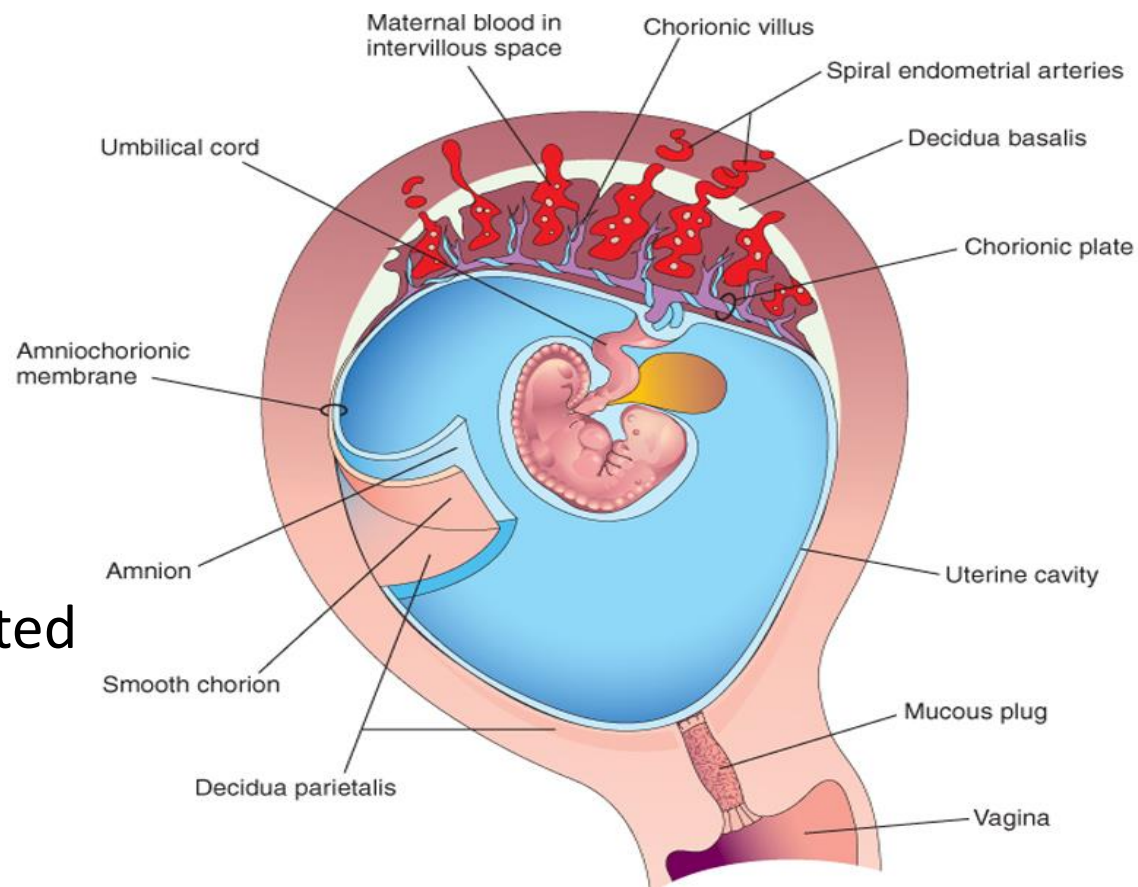


C



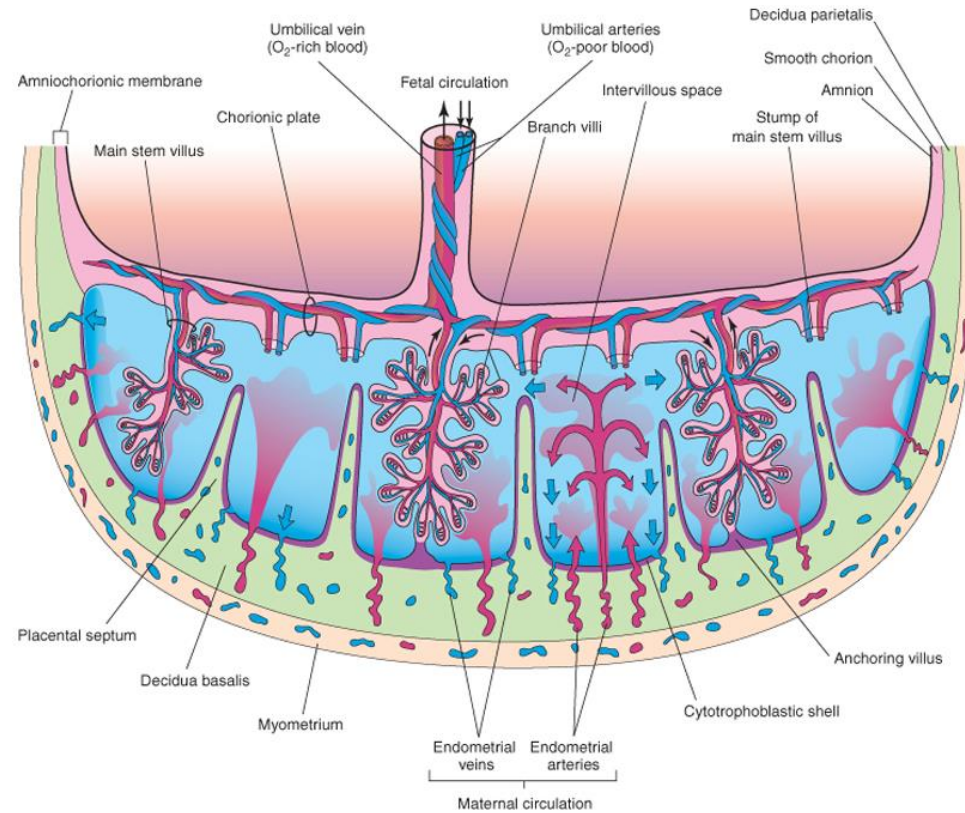
D

- **After disappearance of the decidua capsularis,**
- the smooth part of the chorionic sac fuses with the decidua parietalis
- This fusion can be separated and usually occurs when blood escapes from the intervillous space
- The collection of blood (hematoma) pushes the chorionic membrane away from the decidua parietalis, thereby reestablishing the potential space of the uterine cavity.

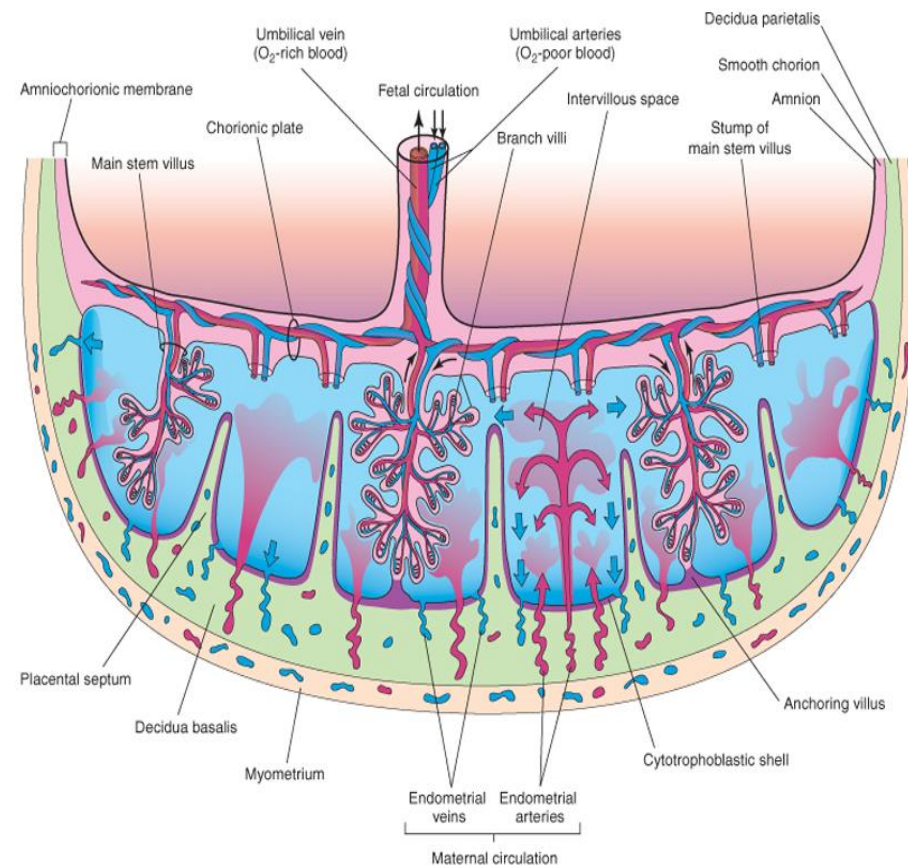
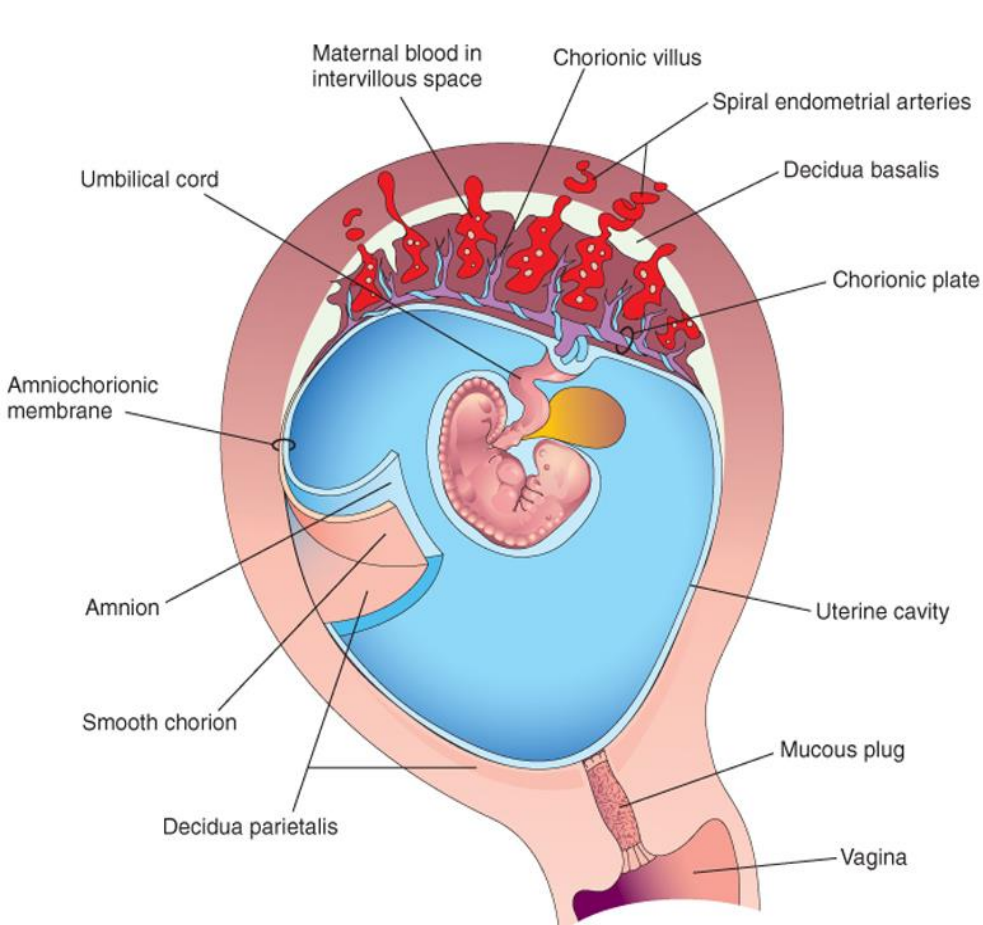


The intervillous space of the placenta,

- contains maternal blood,
- is derived from the lacunae that developed in the syncytiotrophoblast during the second week of development
- is divided into compartments by the **placental septa**, however, there is free communication between the compartments because the septa do not reach the chorionic plate
- This large blood-filled space results from the coalescence and enlargement of the lacunar networks



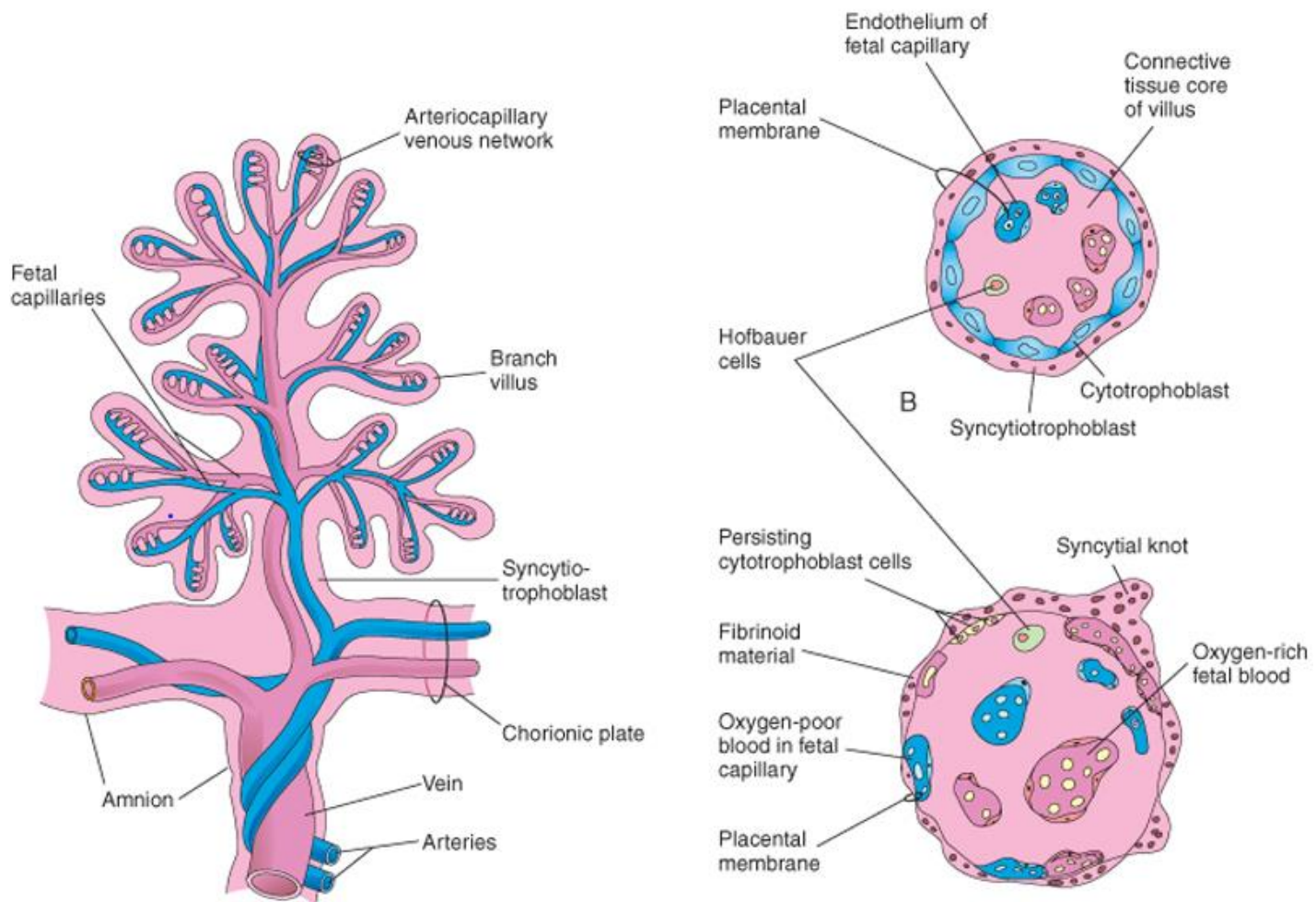
- Maternal blood enters the intervillous space from the **spiral endometrial arteries** in the decidua basalis
- The spiral arteries pass :
 - through gaps in the **cytotrophoblastic shell**
 - discharge blood into the intervillous space.
- This large space is drained by **endometrial veins** that also penetrate the cytotrophoblastic shell
- Endometrial veins are found over the entire surface of the decidua basalis
- The numerous branch chorionic villi:
 - arising from stem villi
 - are continuously showered with maternal blood that circulates through the intervillous space
- The blood in this space carries oxygen and nutritional materials that are necessary for fetal growth and development
- The maternal blood also contains fetal waste products such as carbon dioxide, salts, and products of protein metabolism.



- The amniotic sac enlarges faster than the chorionic sac
- As a result, the amnion and smooth chorion soon fuse to form the **amniochorionic membrane**
- This composite membrane fuses with the decidua capsularis and, after disappearance of this capsular part of the decidua, adheres to the decidua parietalis
- It is the amniochorionic membrane that ruptures during labor (**the expulsion of the fetus and placenta from the uterus**)
- Preterm rupture of this membrane is the most common event leading to premature labor
- When the membrane ruptures, amniotic fluid escapes through the cervix and vagina to the exterior.

Placental Circulation

- The branch chorionic villi of the placenta provide a large surface area where materials may be exchanged across the very thin placental membrane ("barrier") interposed between the fetal and maternal circulations
- It is through the numerous branch villi that arise from the stem villi that the main exchange of material between the mother and fetus takes place
- The circulations of the fetus and the mother are separated by the placental membrane consisting of extrafetal tissues



A Drawing of a stem chorionic villus showing its arteriocapillary-venous system. The arteries carry poorly oxygenated fetal blood and waste products from the fetus, whereas the vein carries oxygenated blood and nutrients to the fetus. **B** and **C**, Drawings of sections through a branch villus at 10 weeks and full term, respectively. The placental membrane, composed of extrafetal tissues, separates the maternal blood in the intervillous space from the fetal blood in the capillaries in the villi. Note that the placental membrane becomes very thin at full term. Hofbauer cells are thought to be phagocytic cells.

Fetal Placental Circulation

- Poorly oxygenated blood leaves the fetus,
- and passes through the umbilical arteries to the placenta
- At the site of attachment of the umbilical cord to the placenta
 - arteries divide into several radially disposed **chorionic arteries** that branch freely in the chorionic plate before entering the chorionic villi
- an extensive arteriocapillary-venous system within the chorionic villi,
 - ✓ brings the fetal blood extremely close to the maternal blood
 - ✓ provides a very large surface area for the exchange of metabolic and gaseous products between the maternal and fetal bloodstreams
 - ✓ no intermingling of fetal and maternal blood;
 - ✓ however, very small amounts of fetal blood may enter the maternal circulation when minute defects develop in the placental membrane
 - ✓ The well-oxygenated fetal blood in the fetal capillaries passes into thin-walled veins that follow the chorionic arteries to the site of attachment of the umbilical cord
- They converge here to form the umbilical vein
- This large vessel carries oxygen-rich blood to the fetus

Maternal Placental Circulation

- The maternal blood in the intervillous space
 - is temporarily outside the maternal circulatory system
 - enters the intervillous space through **80 to 100 spiral endometrial arteries**
 - discharge into the intervillous space through gaps in the **cytotrophoblastic shell**
 - The blood flow from the **spiral arteries** is pulsatile
 - and is propelled in jetlike fountains by the maternal blood pressure
 - The entering blood is at a considerably higher pressure than that in the intervillous
 - As the pressure dissipates, the blood flows slowly over the branch villi, allowing an **exchange of metabolic and gaseous products with the fetal blood**
 - The blood eventually returns through the endometrial veins to the maternal circulation.

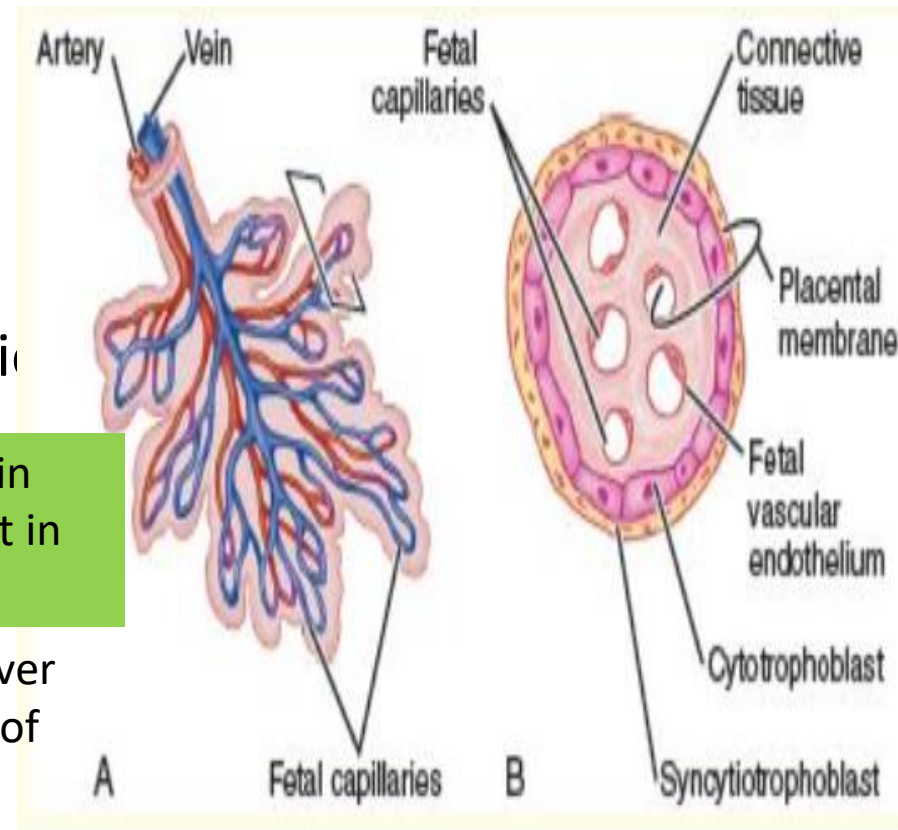
- Reductions of uteroplacental circulation result in **fetal hypoxia** and **intrauterine growth restriction (IUGR)**.
- Severe reductions of uteroplacental circulation may result in **fetal death**
- The intervillous space of the mature placenta contains **≈ 150 mL of blood that is replenished 3 or 4 times per minute**
- The intermittent contractions of the uterus during pregnancy decrease uteroplacental blood flow slightly; however, they do not force significant amounts of blood out of the intervillous space
- Consequently, oxygen transfer to the fetus is decreased during uterine contractions, but does not stop.

The Placental Membrane

- The placental membrane is a composite structure that consists of the extrafetal tissues separating the maternal and fetal blood.
- Until approximately 20 weeks, **the placental membrane** consists of four layers:
 - syncytiotrophoblast
 - cytotrophoblast
 - connective tissue of villus
 - endothelium of fetal capillaries

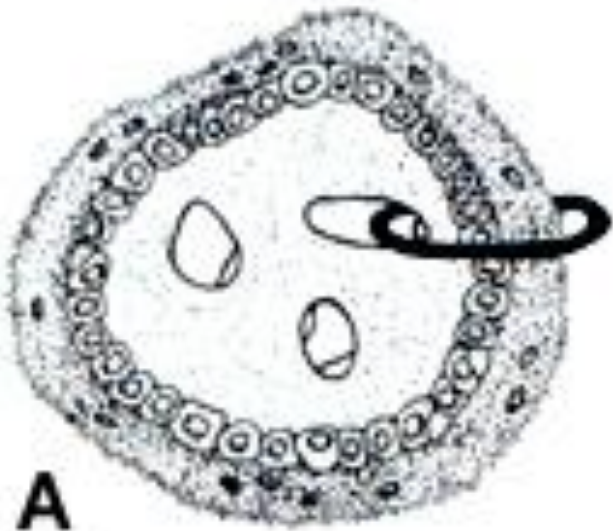
• After the 20th week, histologic changes occur in the branch villi that result in the cytotrophoblast in many of the villi becoming attenuated

• Eventually **cytotrophoblastic cells disappear** over large areas of the villi, leaving only thin patches of **syncytiotrophoblast**



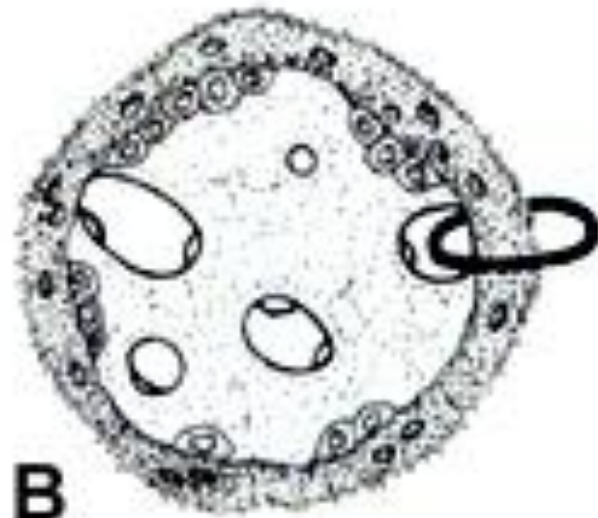
Placental membrane (barrier): formed by

- 1) **Endothelium** and its basal lamina of villous capillaries;
- 2) **Connective tissue** in the core of villi;
- 3) **Cytotrophoblast** and its basal lamina;
- 4) **Syncytiotrophoblast** layer.



A

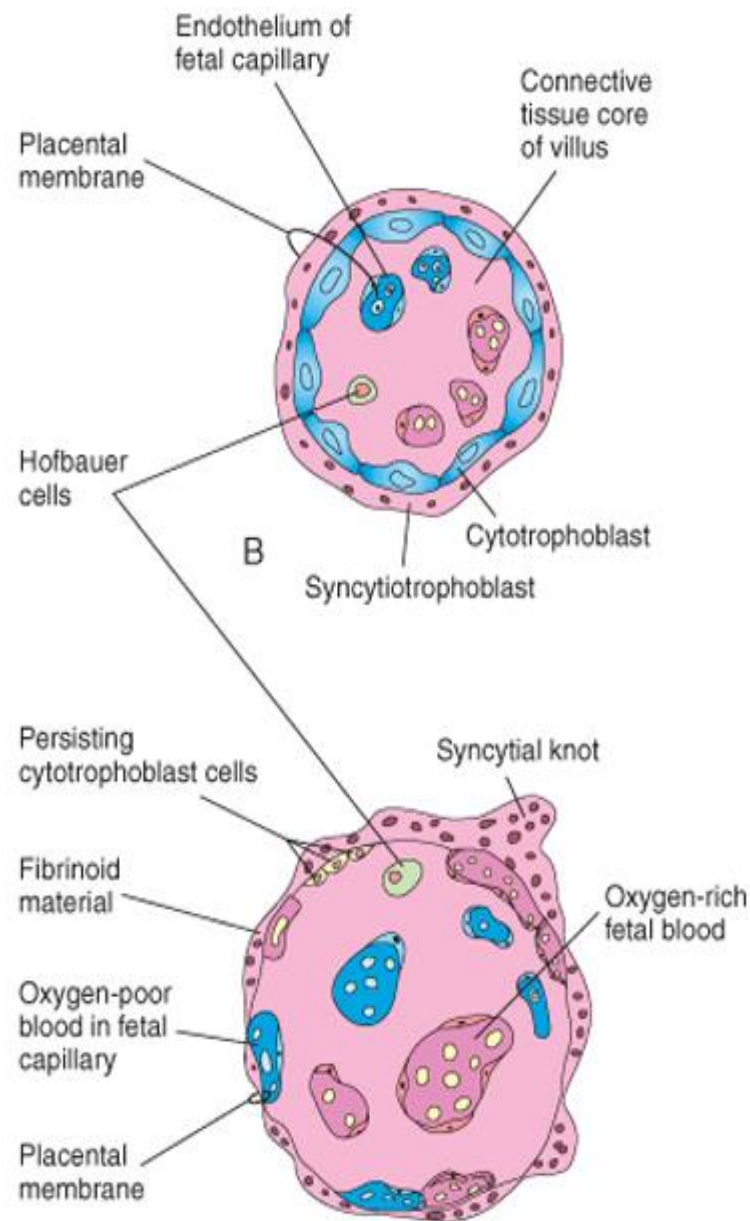
early stage



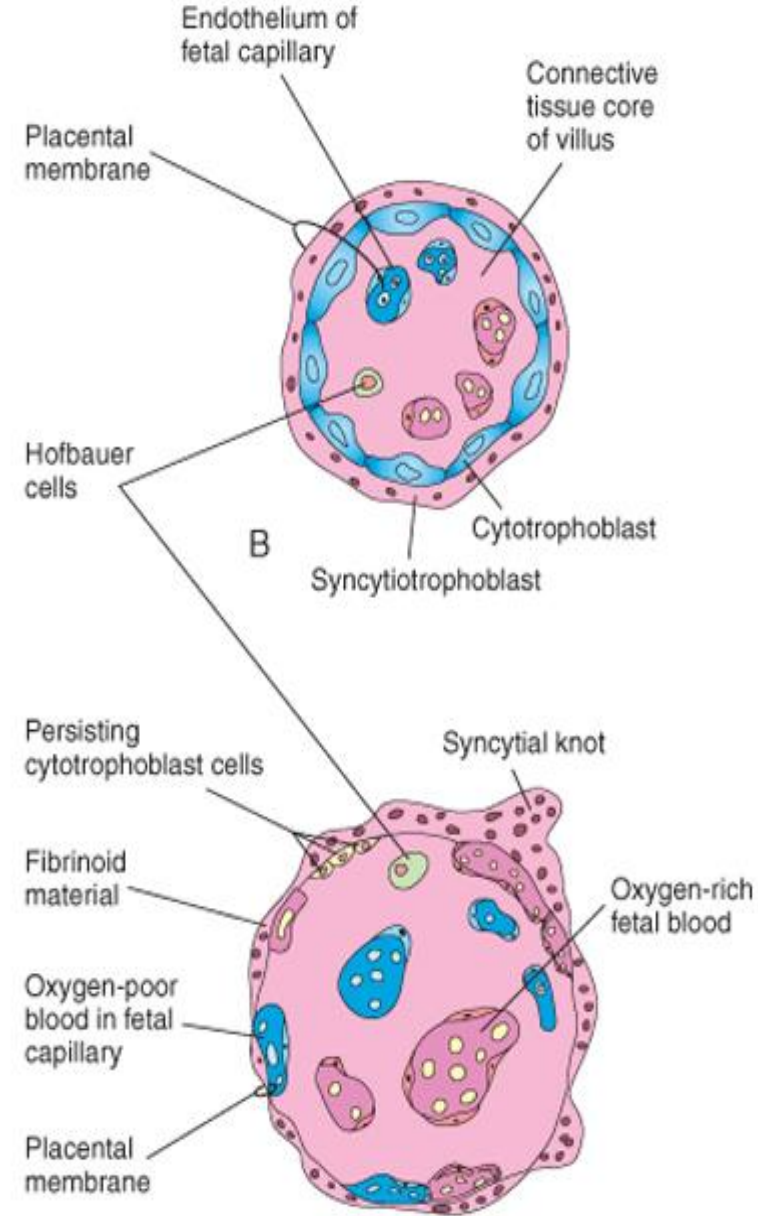
B

after 4th month

- the placental membrane consists of three layers in most places
- In some areas, the placental membrane becomes markedly **thinned and attenuated**.
- At these sites, the syncytiotrophoblast comes in direct contact with the endothelium of the fetal capillaries to form a **vasculosyncytial** placental membrane
- The placental membrane was formerly called the placental barrier, an inappropriate term because there are only a few substances, endogenous or exogenous, that are unable to pass through the placental membrane in detectable amounts.
- The placental membrane acts as a barrier only when the molecule is of a certain size, configuration, and charge such as heparin and bacteria.
- **Some metabolites, toxins, and hormones**, although present in the maternal circulation, **do not pass through the placental membrane** in sufficient concentrations to affect the embryo/fetus.



- Most drugs and other substances in the maternal plasma pass through the placental membrane and enter the fetal plasma
- Electron micrographs of the syncytiotrophoblast show that **its free surface has many microvilli, more than 1 billion/cm² at term**, that increase the surface area for exchange between the maternal and fetal circulations
- As pregnancy advances, the placental membrane becomes progressively thinner so that blood in many fetal capillaries is extremely close to the maternal blood in the intervillous space

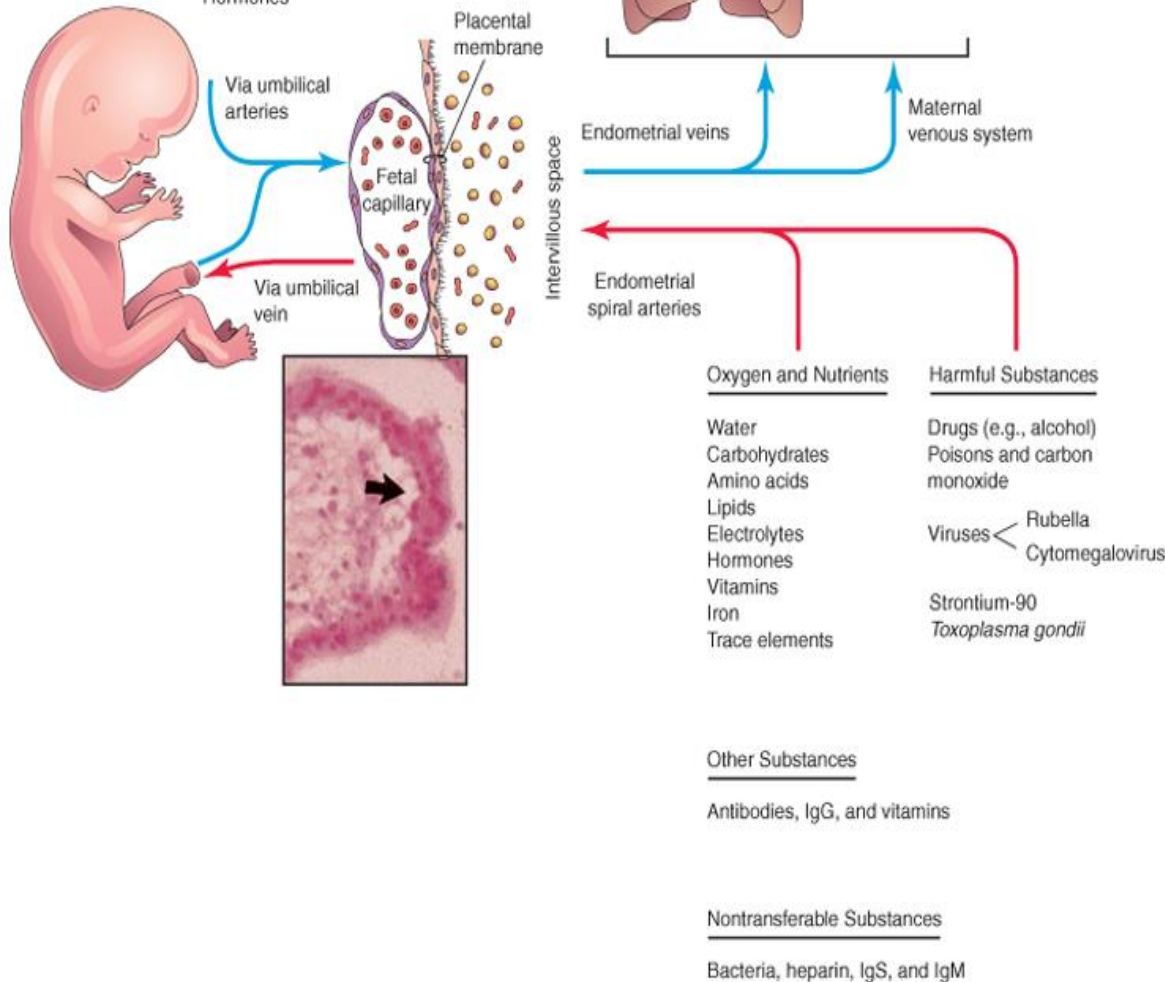


Waste Products

Carbon dioxide, water,
urea, uric acid, bilirubin

Other Substances

RBC antigens
Hormones



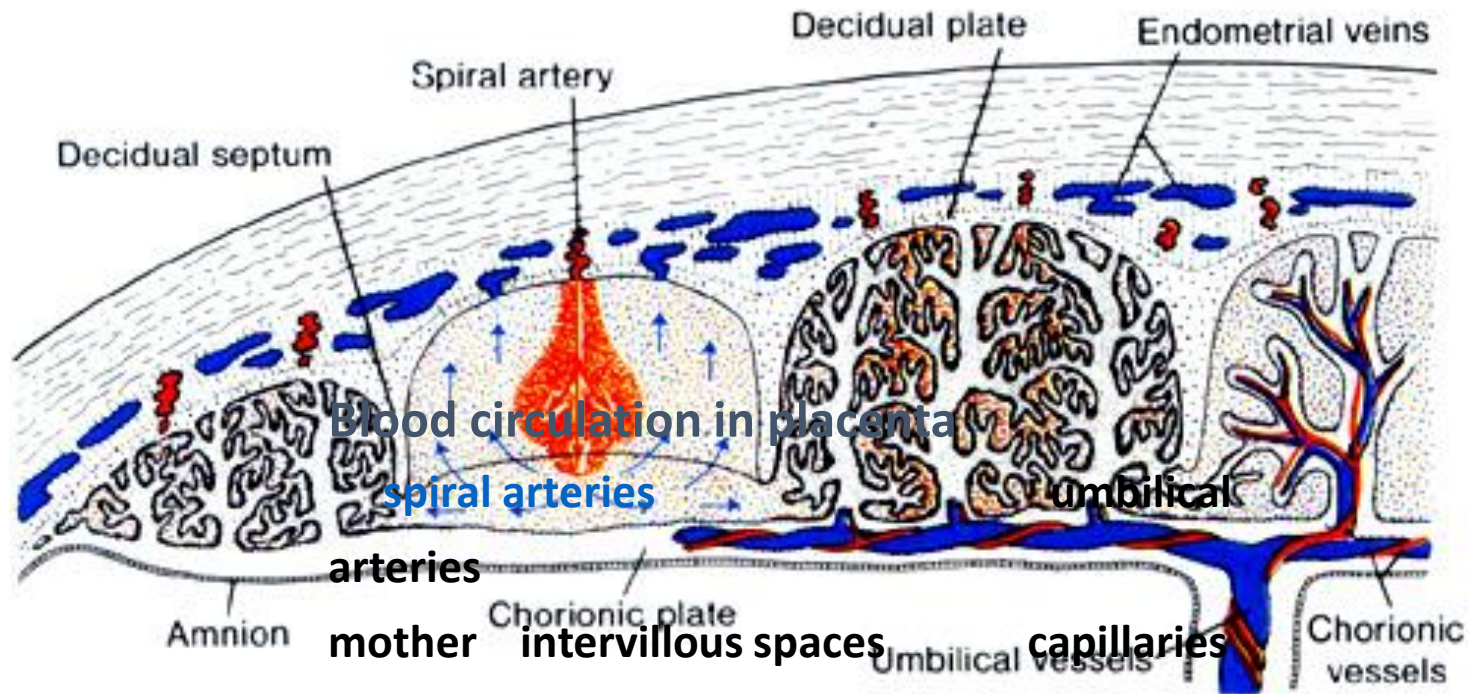
Diagrammatic illustration of **transfer across the placental membrane (barrier)**

The extrafetal tissues, across which transport of substances between the mother and fetus occurs, collectively constitute the placental membrane.

Inset, Light micrograph of chorionic villus showing a fetal capillary (arrow) and the placental membrane.

At full term:

- 1) discoid, d = 15 -20 cm, t = 2.5 cm, w = 500 g
 - 2) fetal side: smooth with the cord attached in the center
 - 3) maternal side: rough with 15 - 30 cotyledons
- Three layer structure:
 - **Chorionic plate** → **stem villi** → **cytotrophoblastic shell**
 - **Basal plate** → **placental septa** → **cotyledons**
 - **Free villi + intervillous spaces** (maternal blood)



Blood circulation in placenta

spiral arteries

umbilical

arteries

mother

intervillous spaces

capillaries

Chorionic vessels

Umbilical vessels

fetus

uterine veins

umbilical vein

Functions of the Placenta

- **Has three main functions:**
 1. Metabolism (e.g., synthesis of glycogen)
 2. Transport of gases and nutrients
 3. Endocrine secretion (e.g., human chorionic gonadotropin (hCG))
- These comprehensive activities are essential for maintaining pregnancy and promoting normal fetal development.

1. Placental Metabolism

- The placenta, particularly during early pregnancy, **synthesizes**:
 - Glycogen
 - Cholesterol
 - fatty acidswhich serve as sources of nutrients and energy for the embryo/fetus.
- Many of its metabolic activities are undoubtedly critical for its other two major placental activities (transport and endocrine secretion).

2. Placental Transfer

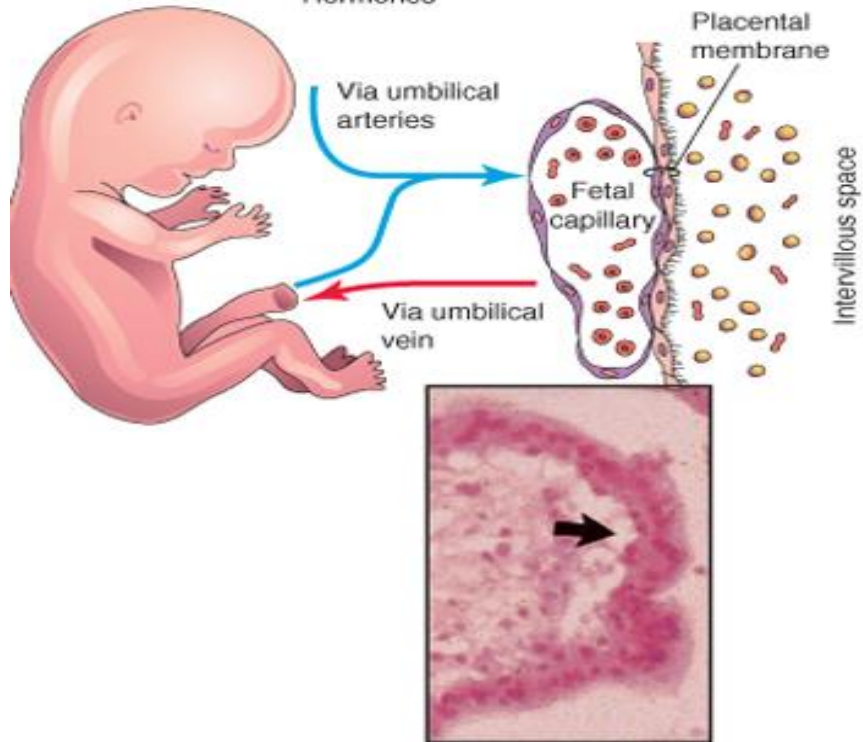
- The transport of substances in both directions between the fetal and maternal blood is facilitated by the great surface area of the placental membrane.
 - Almost all materials are transported across the placental membrane by one of the following **four main transport** mechanisms:
 - simple diffusion
 - facilitated diffusion
 - active transport, and pinocytosis

Waste Products

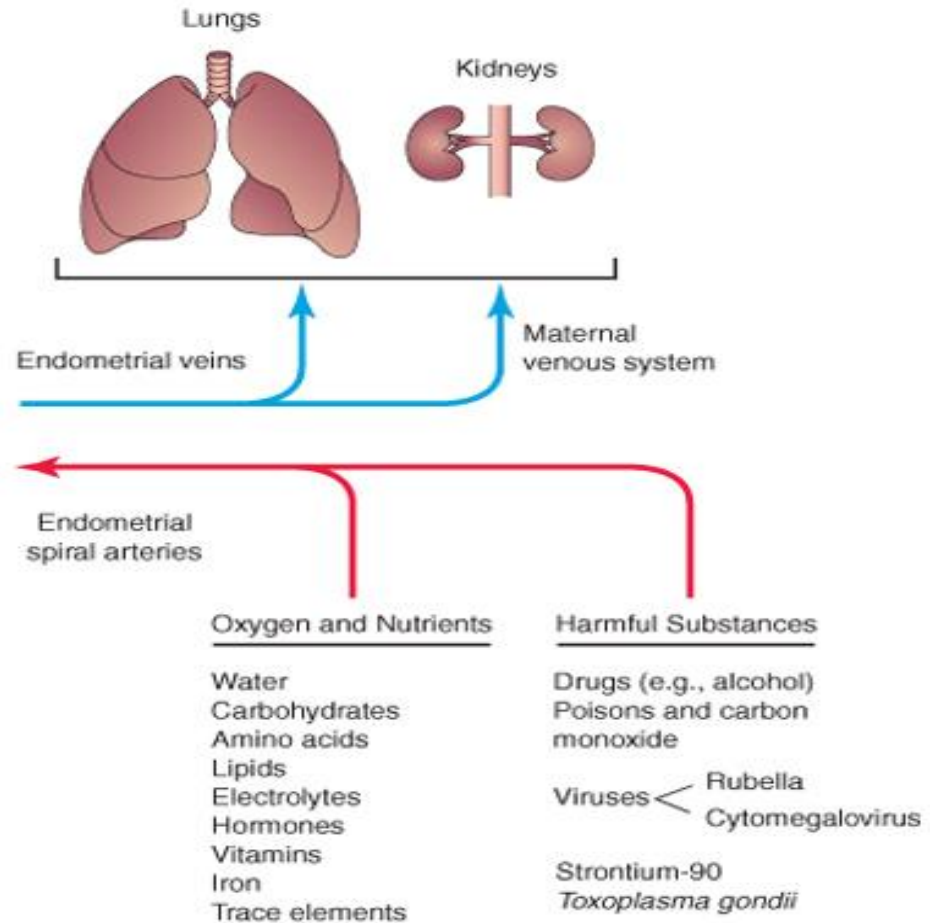
Carbon dioxide, water,
urea, uric acid, bilirubin

Other Substances

RBC antigens
Hormones



Maternal



Other Substances

Antibodies, IgG, and vitamins

Nontransferable Substances

Bacteria, heparin, IgS, and IgM

- **simple diffusion** (passive transport)
 - is usually characteristic of substances moving from areas of higher to lower concentration until equilibrium is established.
- In **facilitated diffusion**:
 - is transport through electrical gradients
- **Active transport**:
 - Is against a concentration gradient and requires energy.
 - Such systems may involve carrier molecules that temporarily combine with the substances to be transported
- **Pinocytosis**
 - is a form of endocytosis in which the material being engulfed is a small amount of extracellular fluid.
 - This method of transport is usually reserved for large molecules.
 - Some proteins are transferred very slowly through the placenta by pinocytosis.

Other Placental Transport Mechanisms

- There are three other methods of transfer across the placental membrane
 - **In the first,**
 - ✓ fetal red blood cells pass into the maternal circulation, particularly during parturition, through microscopic breaks in the placental membrane
 - ✓ Labeled maternal red blood cells have also been found in the fetal circulation
 - ✓ Consequently, red blood cells may pass in either direction through very small defects in the placental membrane
 - **In the second method of transport,**
 - ✓ cells cross the placental membrane under their own power, e.g., maternal leukocytes and *Treponema pallidum*, the organism that causes syphilis
 - **In the third method of transport,**
 - ✓ some bacteria and protozoa such as *Toxoplasma gondii* infect the placenta by creating lesions and then cross the placental membrane through the defects that are created

• **Transfer of Gases**

- Oxygen, carbon dioxide, and carbon monoxide cross the placental membrane by simple diffusion
- Interruption of oxygen transport for several minutes endangers survival of the embryo or fetus
- The placental membrane approaches the efficiency of the lungs for gas exchange
- The quantity of oxygen reaching the fetus is primarily flow limited rather than diffusion limited; hence, fetal hypoxia (decreased levels of oxygen) results primarily from factors that diminish either the uterine blood flow or fetal blood flow.

• **Nutritional Substances**

- **Nutrients** constitute the bulk of substances transferred from the mother to the embryo/fetus
- **Water** is rapidly exchanged by simple diffusion and in increasing amounts as pregnancy advances
- **Glucose** produced by the mother and placenta is quickly transferred to the embryo/fetus by facilitated diffusion
- **There is little or no transfer of maternal:**
 - cholesterol
 - Triglycerides
 - or phospholipids
- Although there is transport of free fatty acids, the amount transferred appears to be relatively small
- **Amino acids** are actively transported across the placental membrane and are essential for fetal growth.
 - For most amino acids, the plasma concentrations in the fetus are higher than in the mother.
- **Vitamins** cross the placental membrane and are essential for normal development
- **Water-soluble vitamins** cross the placental membrane more quickly than fat-soluble ones.

- **Hormones;**

- Protein hormones do not reach the embryo or fetus in significant amounts, except for a slow transfer of **thyroxine and triiodothyronine**
- Unconjugated steroid hormones cross the placental membrane rather freely
- Testosterone and certain synthetic progestins cross the placental membrane and may cause **masculinization of female fetuses in some cases**

- **Electrolytes**

- are freely exchanged across the placental membrane in significant quantities, each at its own rate
- When a mother receives intravenous fluids with electrolytes, they also pass to the fetus and affect its water and electrolyte status.

- **Maternal Antibodies**

- The fetus produces only small amounts of antibodies because of its immature immune system
- Some passive immunity is conferred on the fetus by the placental transfer of maternal antibodies
- IgG gamma globulins are readily transported to the fetus by transcytosis

- **Maternal antibodies**

- **Grant fetal immunity** to some diseases such as diphtheria, smallpox, and measles; however, no immunity is acquired to pertussis (whooping cough) or varicella (chickenpox)
- **A maternal protein**, transferrin, crosses the placental membrane and carries iron to the embryo or fetus
- The placental surface contains special receptors for this protein.

- ***Hemolytic Disease of the Newborn***

- *Small amounts of fetal blood may pass to the maternal blood through microscopic breaks in the placental membrane*
- *If the fetus is **Rh positive** and the mother **Rh negative**, the fetal blood cells may stimulate the formation of **anti-Rh antibodies** by the immune system of the mother*
- *These pass to the fetal blood and causes hemolysis of fetal Rh-positive blood cells, jaundice, and anemia in the fetus*
- *Some fetuses with hemolytic disease of the newborn, or fetal erythroblastosis, fail to make a satisfactory intrauterine adjustment*
- *They may die unless delivered early or given intrauterine, intraperitoneal, or intravenous transfusions of packed Rh-negative blood cells to maintain the fetus until after birth*
- *Hemolytic disease of the newborn is relatively uncommon now because Rh (D) immunoglobulin given to the mother usually prevents development of this disease in the fetus*

- Waste Products

- Urea and uric acid pass through the placental membrane by simple diffusion
- Conjugated bilirubin (which is fat soluble) is easily transported by the placenta for rapid clearance

- **Drugs and Drug Metabolites**

- Most drugs and drug metabolites cross the placenta by simple diffusion,
- the exception being those with a structural similarity to amino acids, such as methyldopa and antimetabolites
- Some drugs cause major congenital anomalies
- Fetal drug addiction may occur after maternal use of drugs such as heroin and 50% to 75% of these newborns experience withdrawal symptoms.

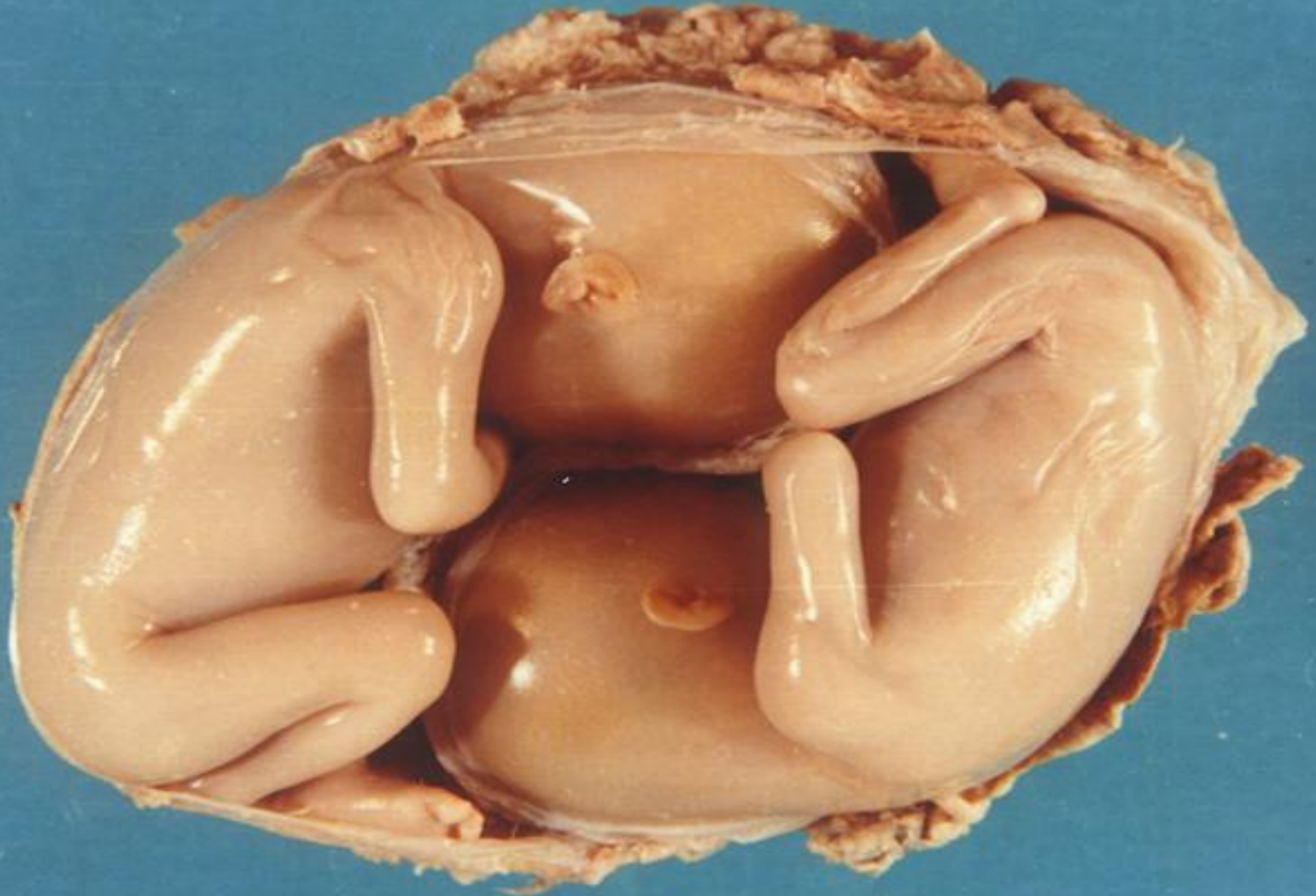
- **Infectious Agents**

- **Cytomegalovirus**, rubella, and coxsackie viruses, and viruses associated with variola, varicella, measles, and poliomyelitis may pass through the placental membrane and cause fetal infection
- In some cases, such as the rubella virus, congenital anomalies, such as cataracts, may be produced
- Microorganisms such as *Treponema pallidum*, which causes syphilis, and *Toxoplasma gondii*, which produces destructive changes in the brain and eyes, also cross the placental membrane, often causing congenital anomalies and/or death of the embryo or fetus

- Placental Endocrine Synthesis and Secretion
 - Using precursors derived from the fetus and/or the mother, the syncytiotrophoblast of the placenta synthesizes protein and steroid hormones
 - The protein hormones synthesized by the placenta are hCG
 - Human chorionic somatomammotropin or human placental lactogen
 - Human chorionic thyrotropin
 - Human chorionic corticotropin

- **Hormone production**
- **Human chorionic gonadotropin (HCG)**: detectable in the blood and urine as early as the **3rd week** and useful for early diagnosis of pregnancy;
- **Human placental lactogen (HPL)**: stimulating growth of breasts in preparation for lactation.
- **Progesterone** and **estrogen**: maintaining pregnancy when corpus luteum degenerates;

Multiple pregnancy

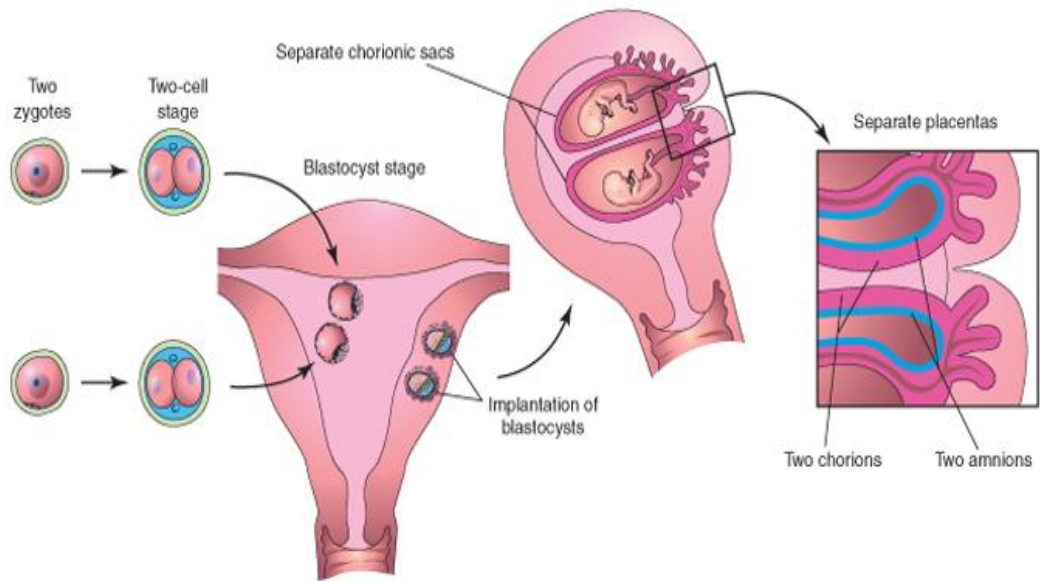


• MULTIPLE PREGNANCIES

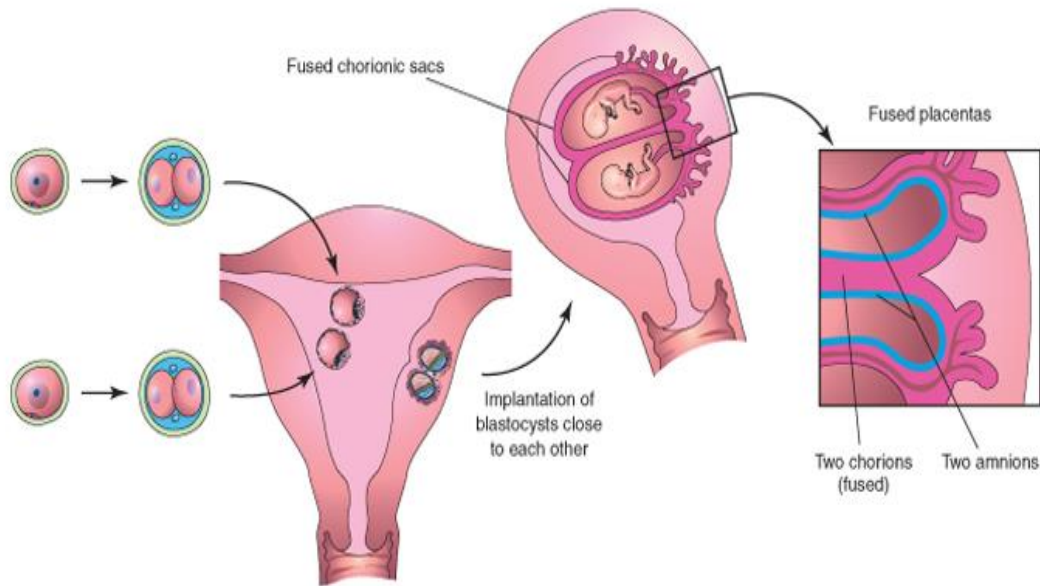
- The risks of chromosomal anomalies
- and fetal morbidity and mortality are higher in multiple gestations than in single gestations
- As the number of fetuses increases, the risks are progressively greater
- In most countries,
 - multiple births are more common now because of greater access to fertility therapies, including induction of ovulation that occurs when exogenous gonadotropins are administered to women with ovulatory failure and to those being treated for infertility by assisted reproductive technologies
 - **In North America**, twins normally occur approximately **once in every 85 pregnancies**, triplets approximately **once in 90² pregnancies**, quadruplets **once in 90³ pregnancies**, and quintuplets approximately once in every 90⁴ pregnancies.

Twins and Fetal Membranes

- Twins that originate from two zygotes are **dizygotic** (DZ) twins or fraternal twins
- whereas twins that originate from one zygote are **monozygotic** (MZ) twins or identical twins
- The fetal membranes and placentas vary according to the origin of the twins
- In the case of MZ twins, the type of placenta and membranes formed depends **on when the twinning process occurs**.
- Approximately **two thirds of twins are DZ**.
- The frequency of **DZ** twinning shows **marked racial differences**, but the incidence of **MZ** twinning is approximately the same in all populations.
- In addition, **the rate of MZ twinning shows** little variation with the mother's age, whereas the rate of **DZ twinning increases with maternal age**.



A



B

Figure 7-26 Diagrams illustrating how **dizygotic twins develop** from two zygotes. The relationships of the fetal membranes and placentas are shown for instances in which the blastocysts implant separately (**A**) and the blastocysts implant close together (**B**).

- In both cases, there are two amnions and two chorions.
- The placentas are usually fused when they implant close together.

- **The study of twins is important in human genetics,**
 - because it is useful for comparing the effects of genes and environment on development
 - If an abnormal condition does not show a simple genetic pattern, comparison of its incidence in MZ and DZ twins may reveal that heredity is involved
 - The tendency for DZ, but not MZ, twins to repeat in families is evidence of hereditary influence.
 - **Studies in a Mormon population showed that the genotype of the mother affects the frequency of DZ twins,**
 - but the genotype of the father has no effect
 - It has also been observed that if the firstborn are twins, a repetition of twinning or some other form of multiple birth is approximately five times more likely to occur with the next pregnancy than in the general population.

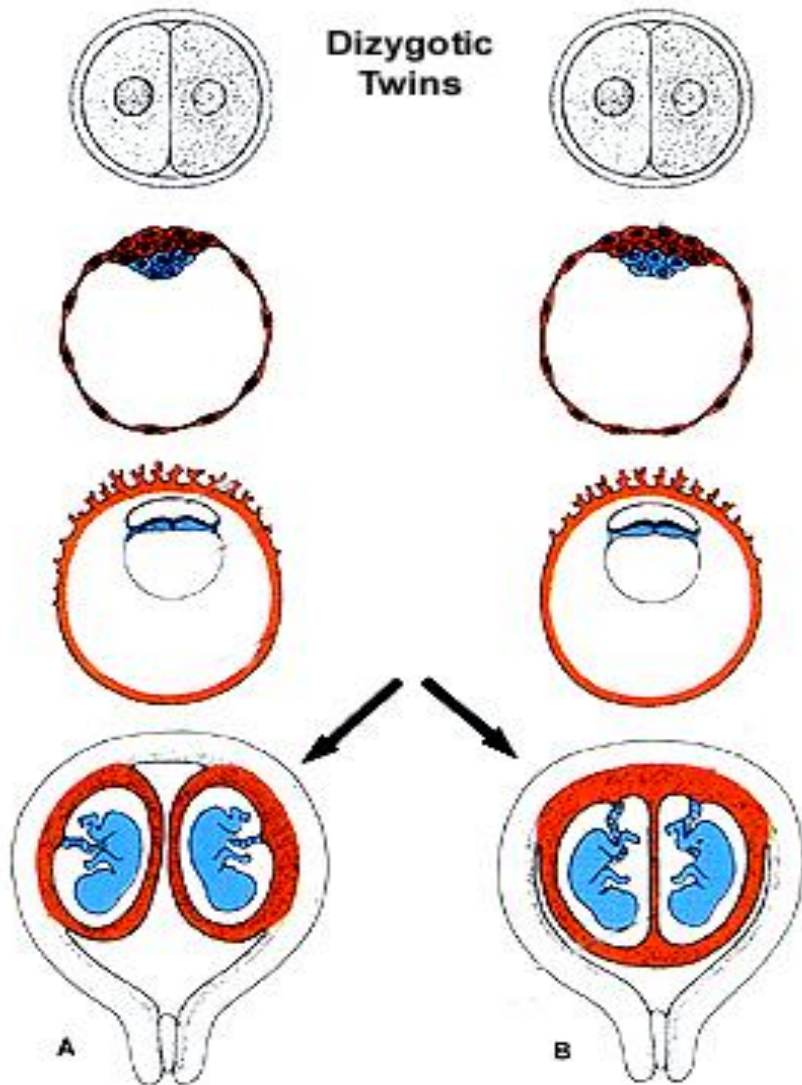
Table 7-1. Frequency of Types of Placentas and Fetal Membranes in Monozygotic (MZ) and Dizygotic (DZ) Twins

ZYGOSITY	SINGLE CHORION		TWO CHORIONS	
	SINGLE AMNION	TWO AMNIONS	FUSED PLACENTAS*	TWO PLACENTAS
MZ	Very rare	65%	25%	10%
DZ	-	-	40% ^{â€}	60%

1. Dizygotic twins

- result from fertilization of two oocytes
- develop from two zygotes and may be of the same sex or different sexes
- For the same reason, they are no more alike genetically than brothers or sisters born at different times
- The only thing they have in common is that they were in their mother's uterus at the same time (i.e., "womb mates")
- DZ twins always:
 - have two amnions and two chorions,
 - but the chorions and placentas may be fused
 - DZ twinning shows a hereditary tendency
 - Recurrence in families is approximately three times that of the general population
 - **The incidence of DZ twinning shows considerable variation, being approximately 1 in 500 in Asians, 1 in 125 in whites, and as high as 1 in 20 in some African populations.**

- **Dizygotic twins:** simultaneous fertilization of two different oocytes by two different sperms



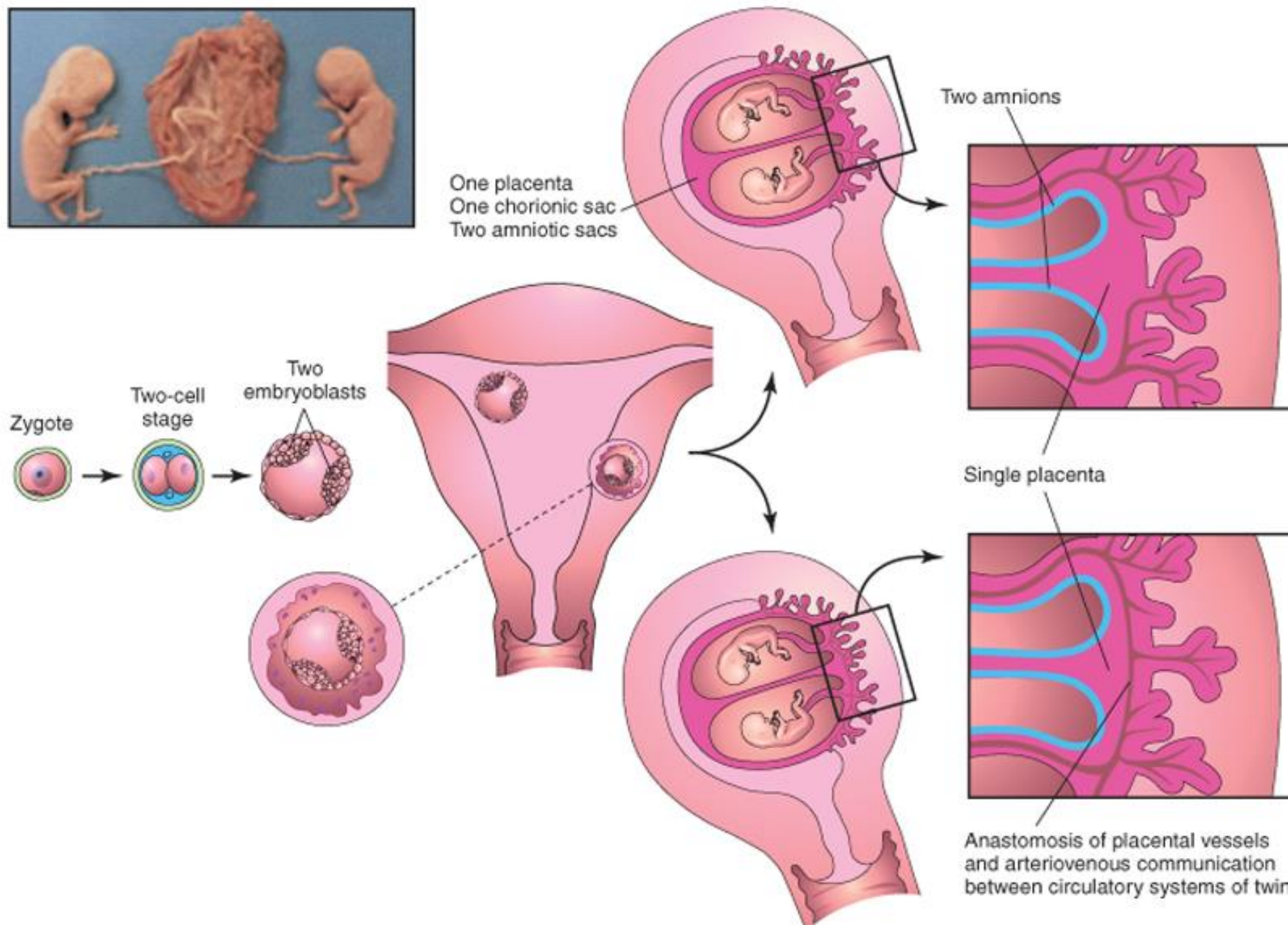
- Separate placenta, chorionic sac and amniotic cavity
- Sometimes two placentas fuse into one.
- Same or different sex;
- External and genetic features are no more alike than other brothers or sisters.

- **Anastomosis of Placental Blood Vessels**

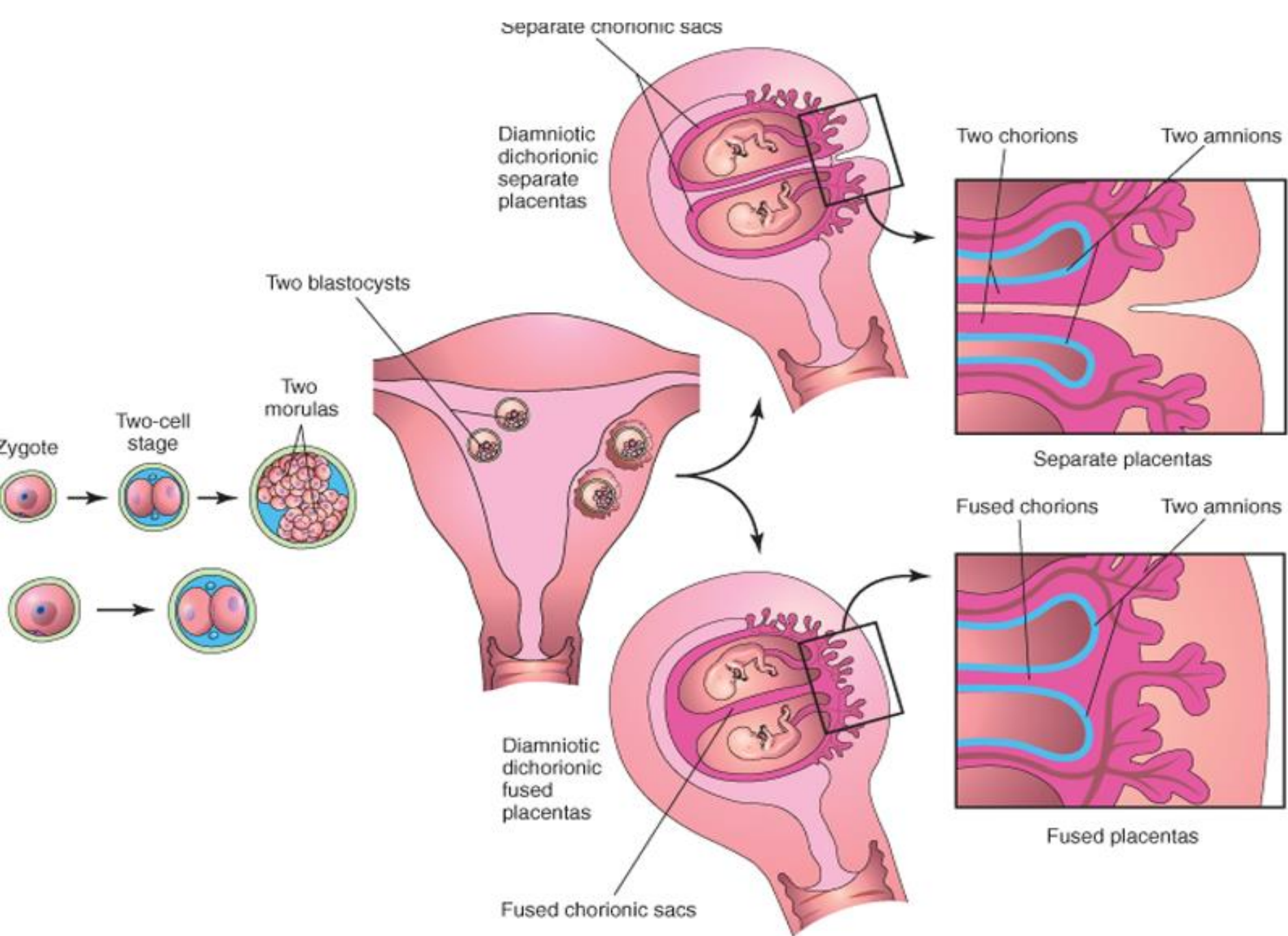
- Anastomoses between blood vessels of fused placentas of DZ twins may result in **erythrocyte mosaicism**.
- **The members of these DZ twins** have red blood cells of two different types because red cells were exchanged between the circulations of the twins
- In cases in which one fetus is a male and the other is female, **masculinization** of the female fetus does not occur

Monozygotic Twins

- result from the fertilization of one oocyte
- develop from one zygote
- **MZ twins**
 - are of the same sex
 - genetically identical
 - and very similar in physical appearance.
 - Physical differences between MZ twins are environmentally induced, e.g., because of anastomosis of placental vessels
- usually begins in the **blastocyst stage**, approximately at the end of the **first week**, and results from division of the embryoblast into two embryonic primordia.
- Subsequently, two embryos, each in **its own amniotic sac**, develop within the **same chorionic sac** and share a common placenta-a monochorionic-diamniotic twin placenta.
-



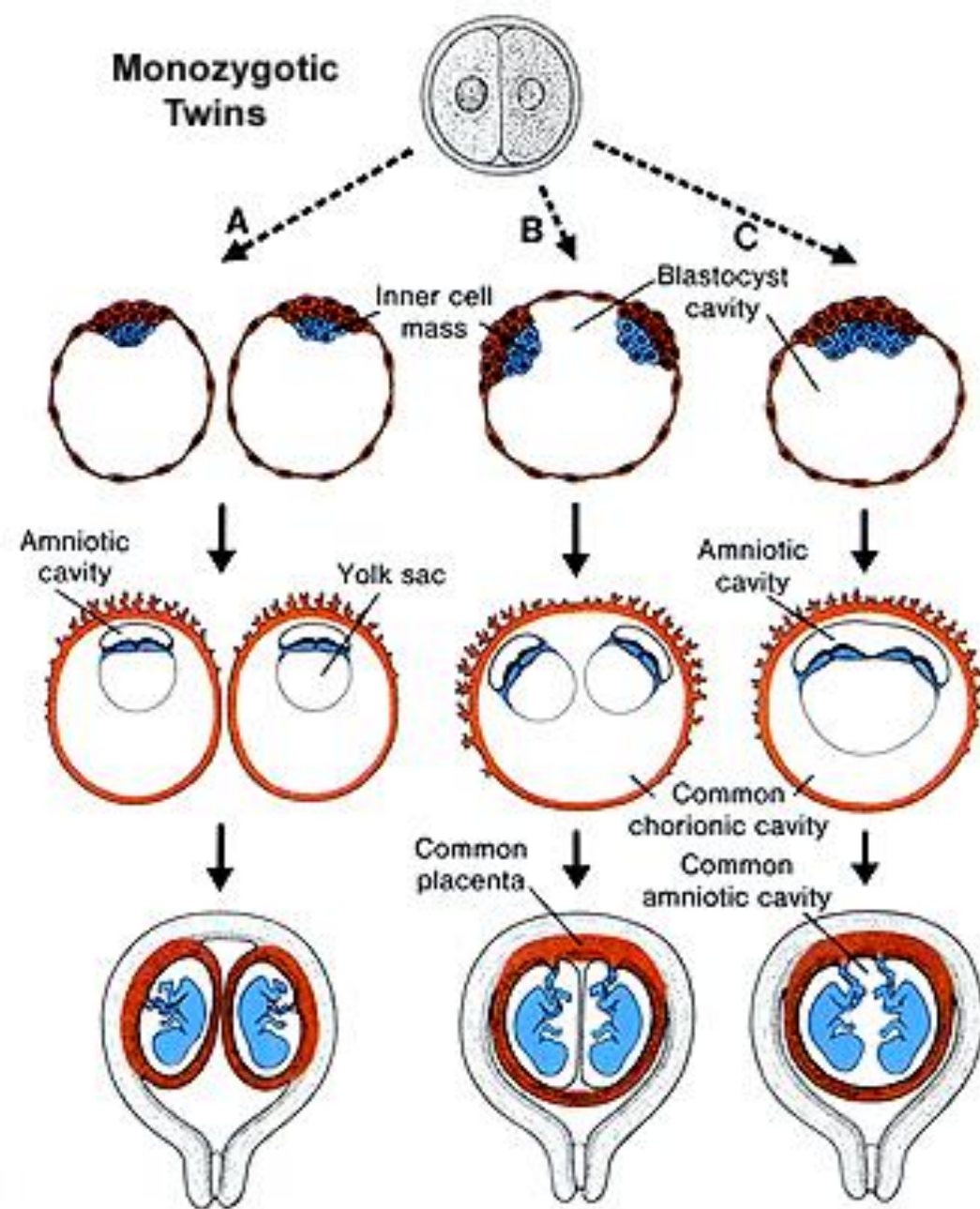
- Uncommonly, early separation of embryonic blastomeres (e.g., during the two- to eight-cell stages) results in MZ twins with **two amnions, two chorions**, and **two placentas** that may or may not be fused
- In such cases, it is impossible to determine from the membranes alone whether the twins are MZ or DZ.



Monozygotic twins:

- resulting from a single fertilized ovum
- Forming at various stages
 - splitting a cleavage stage → **two blastocysts**
 - splitting at stage of early blastocyst → **two inner cell masses**
 - splitting at stage of bilaminar germ disc → **two primitive streaks**
 - same sex, very similar physical appearance
 - Identical blood groups and genetic makeup.

Monozygotic Twins



Establishing the Zygosity of Twins

- Establishing the zygosity of twins is important in tissue and organ transplantation (e.g., bone marrow transplantations)
- The determination of twin zygosity is now done by molecular diagnosis because any two people who are not MZ twins are virtually certain to show differences in some of the large number of DNA markers that can be studied.
- Approximately 35% of MZ twins result from early separation of the embryonic blastomeres, i.e., during the first 3 days of development

- **The other 65% of MZ twins** originate at the **end of the first week of development**
- **Late division of early embryonic cells**, such as division of the embryonic disc during the second week, **results in MZ twins** that are in **one amniotic sac and one chorionic sac**
- A monochorionic-monoamniotic twin placenta is associated with a fetal mortality rate approaching 50%
- These MZ twins are rarely delivered alive because the umbilical cords are frequently so entangled that circulation of blood through their vessels ceases and one or both fetuses die.
- Sonography plays an important role in the diagnosis and management of twin pregnancies
- Ultrasound evaluation is necessary to identify various conditions that may complicate MZ twinning such as IUGR, fetal distress, and premature labor.

Early Death of a Twin

- Because **ultrasonographic studies are a common part of prenatal care**, it is known that early death and resorption of one member of a **twin pair is fairly common**
- Awareness of this possibility must be considered when discrepancies occur between prenatal cytogenetic findings and the karyotype of an infant
- Errors in prenatal cytogenetic diagnosis may arise if extraembryonic tissues (e.g., part of a chorionic villus) from the resorbed twin are examined.

Superfecundation

- Superfecundation is the fertilization of two or more oocytes at different times.
- In humans, the presence of two fetuses in the uterus caused by fertilization at different times (superfetation) is rare.
- DZ human twins with different fathers have been confirmed by genetic markers.

Conjoined Monozygotic

- Twins If the **embryonic disc does not divide** completely, or adjacent embryonic discs fuse, various types of **conjoined MZ twins may form**.
- These **attached** (Greek, *pagos*, fixed) twins are named according to the regions that are attached, e.g., **thoracopagus** indicates that there is anterior union of the thoracic regions.
- It has been estimated that the incidence of conjoined twins is **1 in 50,000 to 100,000 births**.
- In some cases, the twins are connected to each other by skin only or by cutaneous and other tissues, e.g., fused livers .
- Some conjoined twins can be successfully separated by surgical procedures; however, the anatomic relations in most conjoined twins do not permit surgical separation with sustained viability

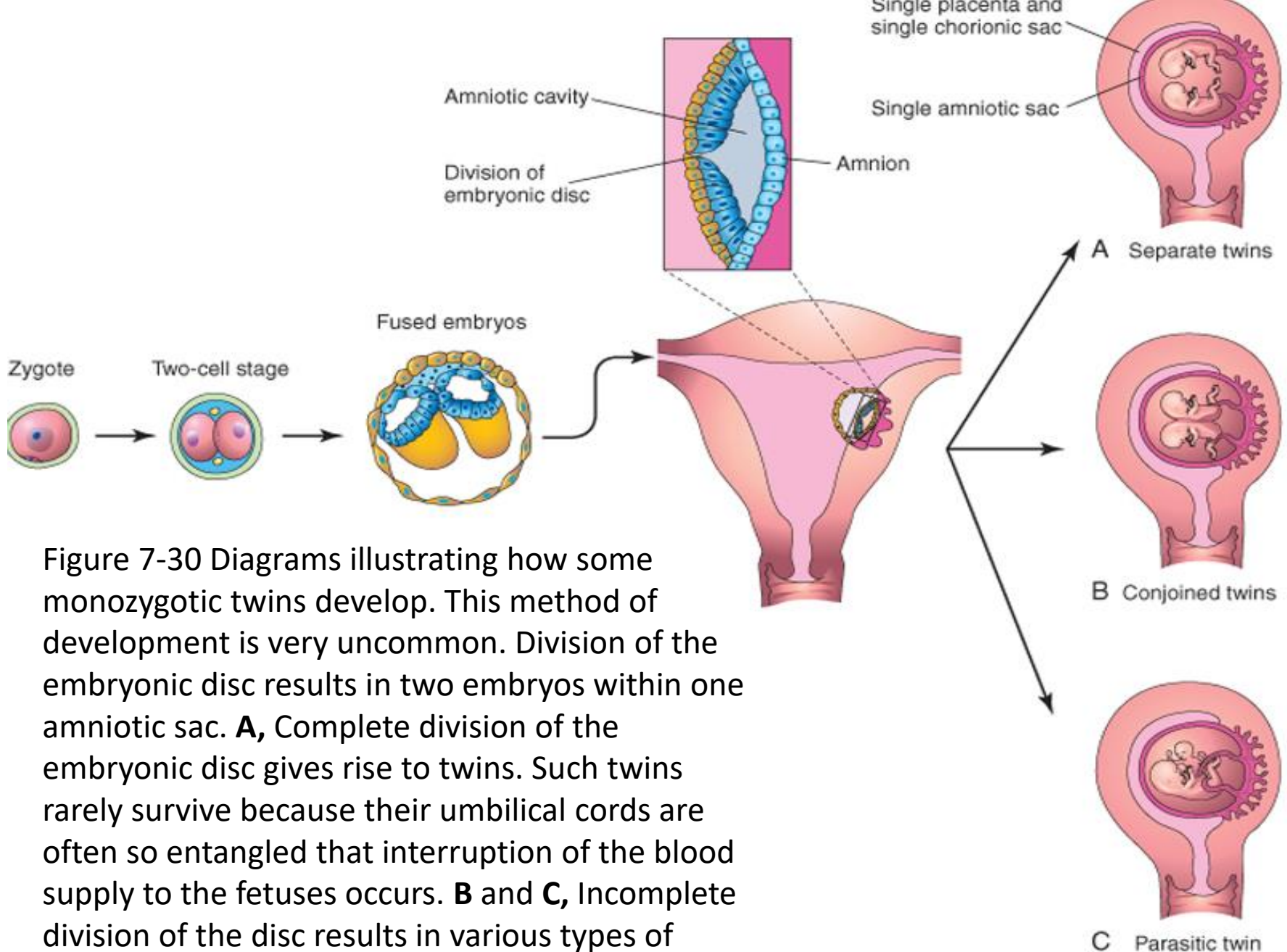


Figure 7-30 Diagrams illustrating how some monozygotic twins develop. This method of development is very uncommon. Division of the embryonic disc results in two embryos within one amniotic sac. **A**, Complete division of the embryonic disc gives rise to twins. Such twins rarely survive because their umbilical cords are often so entangled that interruption of the blood supply to the fetuses occurs. **B** and **C**, Incomplete division of the disc results in various types of conjoined twins.

- The fetal membranes and placenta(s) in multiple pregnancies vary considerably, depending on:
 - the derivation of the embryos
 - and the time when division of embryonic cells occurs.
- **The common type of twins is DZ twins:**
 - with two amnions
 - two chorions and two placentas that may or may not be fused.
- **MZ twins, the less common type:**
 - represent approximately one third of all twins
 - they are derived from one zygote.
 - MZ twins commonly have:
 - one chorion, two amnions, and one placenta. Twins with one amnion, one chorion, and one placenta are always monozygotic, and their umbilical cords are often entangled

Conjoined twins



thoracopagus



omphalopagus



dicephalus



**asymmetrical
duplicity**

- **Other Types of Multiple Births**

- **Triplets may be derived from:**

- One zygote and be identical
 - Two zygotes and consist of identical twins and a singleton
 - Three zygotes and be of the same sex or of different sexes

Thank You!